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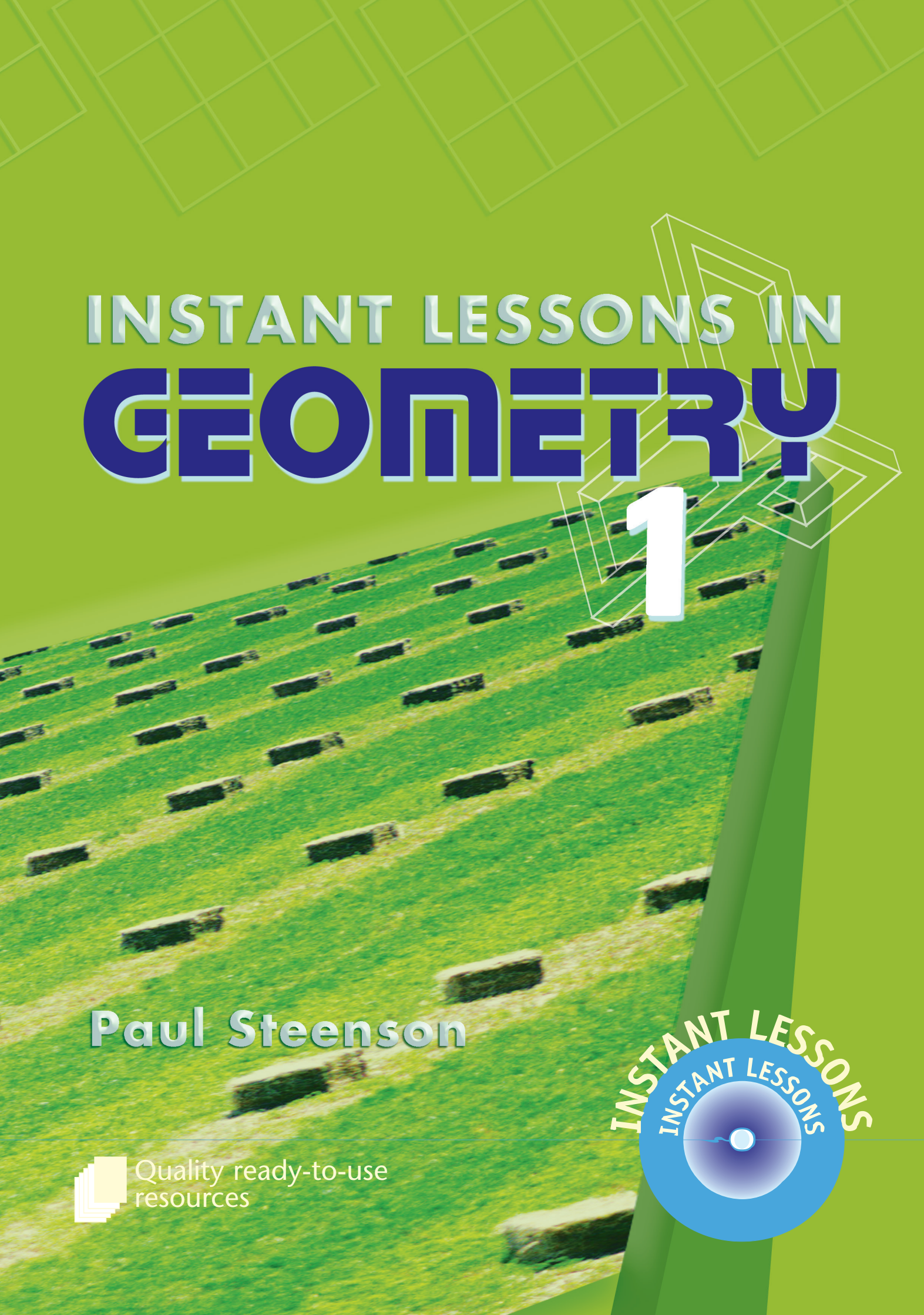

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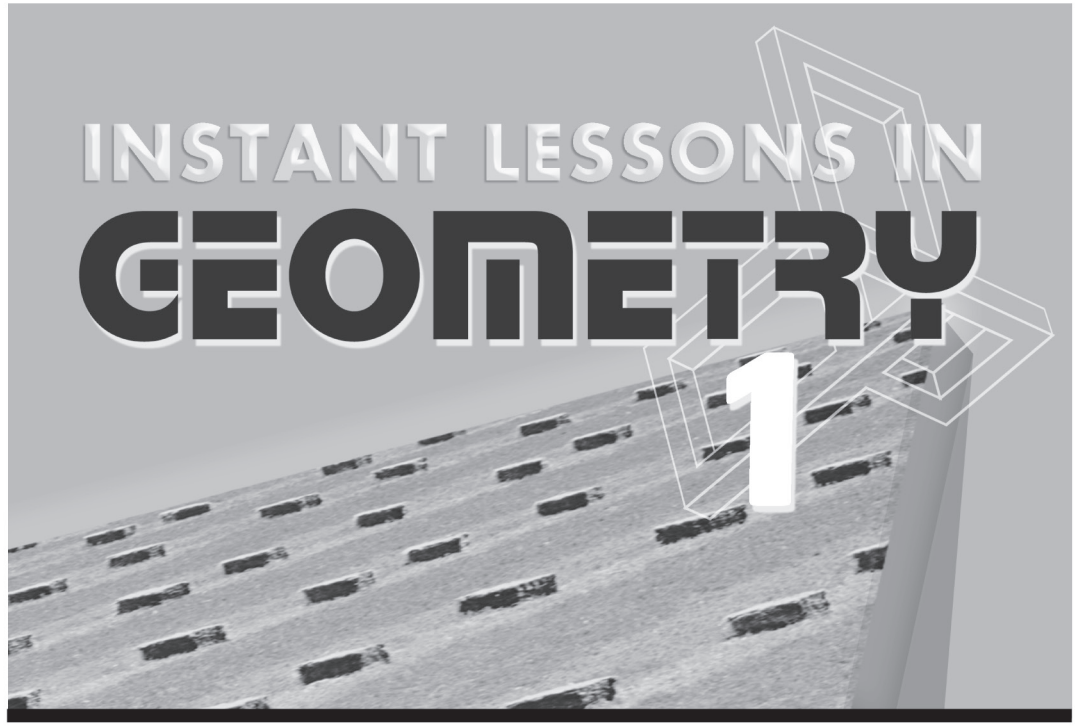

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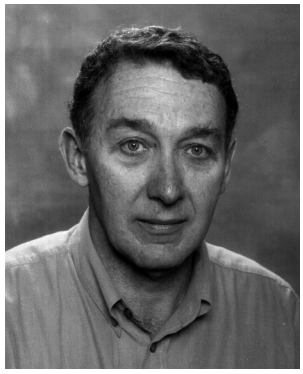

Paul Steenson has more than thirty years' experience teaching Mathematics in secondary schools in Australia. He has been Head of Department since 1984 in a variety of schools in NSW. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a marker of both School Certificate and HSC examination papers. He has been a regular presenter of curriculum workshops and teacher development activities. He has been a co-author of textbooks and written a number of titles in the *Instant Lessons* series.

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Introduction

This book provides a systematic and comprehensive coverage of selected topics from the junior mathematics curriculum. It is independent of any textbook and is not linked to any syllabus. Further geometry topics can be found in *Instant Lessons in Geometry Book 2*.

Most worksheets follow a format comprising three sections:

- **Section A** consists of a set of introductory questions or review of previous knowledge, which can be used as a lesson starter or 'warm up'.
- **Section B** provides an investigation, skills development or drill, with graded questions in the focus area of the worksheet.
- **Section C** provides further practice in the form of different types of questions, challenge or extension questions.

Review worksheets are also included in each topic area.

There are minimal examples or explanations as teachers have different methods of working that they would like their students to follow. Most worksheets could be used by students in a range of year levels or courses. The teacher may decide to use only one or two sections from a worksheet in a lesson, or as a homework assignment. The worksheets are designed to be flexible to meet the needs of the teacher and students.

A time required for students to complete each worksheet is not suggested as this will vary significantly depending on the year level of the students and their level of mathematical skill. The teacher is the best person to make this judgement.

Suggested answers have been provided for most questions, including some of the guided proofs. In general, answers have not been provided for practical tasks, constructions, sketch diagrams, proofs or literacy activities.

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Angles – basic properties 1.1

A

Types of angles

An *acute angle* measures less than 90° (smaller than a right angle).
A *right angle* measures exactly 90° (a quarter of 1 revolution).
An *obtuse angle* measures more than 90° and less than 180°.
A *straight angle* measures exactly 180° (half of 1 revolution).
A *reflex angle* measures more than 180° and less than 360°.
A *revolution* measures exactly 360°.

What type of angle (acute, right, obtuse, straight or reflex) is shown in each diagram?

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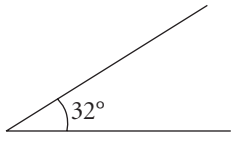

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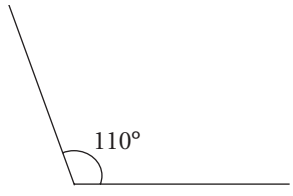

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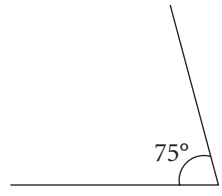

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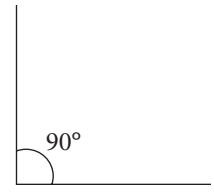

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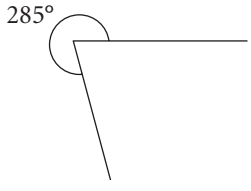

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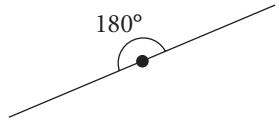

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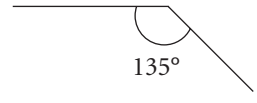

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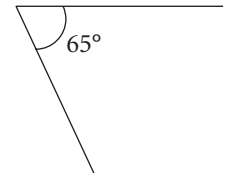


B

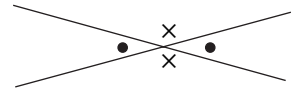
Angle relationships

Complementary angles add to 90°
(e.g. 30° is the complement of 60°).

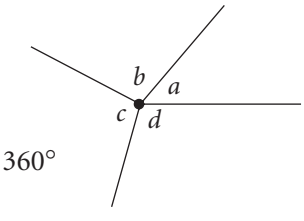
Supplementary angles add to 180°
(e.g. 110° is the supplement of 70°).

Adjacent angles share a common arm and a common vertex.

Vertically opposite angles are the pairs of equal angles formed when two straight lines intersect.



Angles at a point, that form a complete revolution, add to give 360°.



$a + b + c + d = 360°$

Are the following pairs of angles complementary? Answer yes or no.

1 40° and 50° **2** 20° and 70° **3** 65° and 35° **4** 28° and 62°

Write the complement of each of the following angles.

5 60° **6** 55° **7** 75° **8** 32°

Are the following pairs of angles supplementary? Answer yes or no.

9 60° and 120° **10** 150° and 30° **11** 15° and 145° **12** 105° and 75°

Write the supplement of each of the following angles.

13 100° **14** 20° **15** 95° **16** 118°

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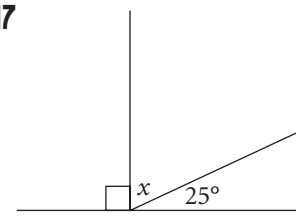
Find the size of the angle marked x in each diagram. (Note: A small square drawn in an angle ⌞ indicates that it is a right angle.)

17



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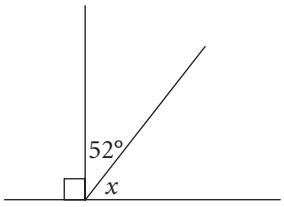

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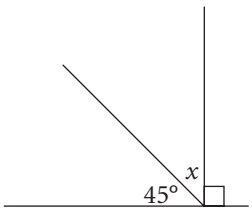

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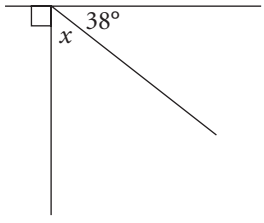

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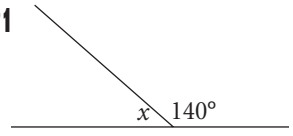

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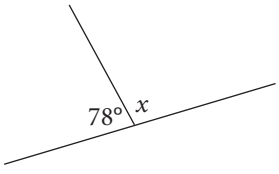

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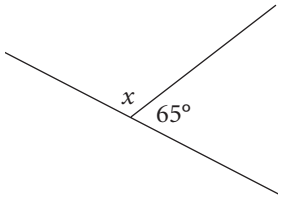

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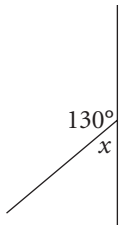

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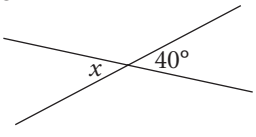

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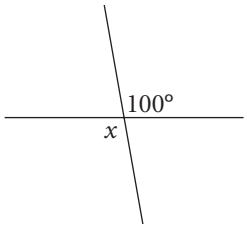

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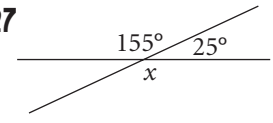

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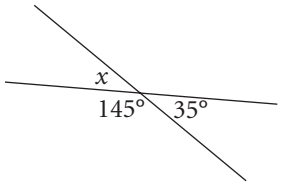

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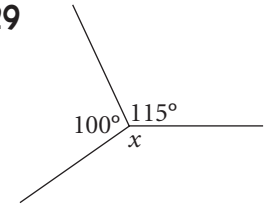

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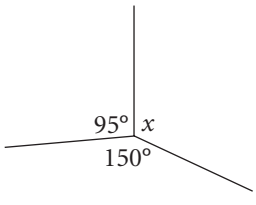

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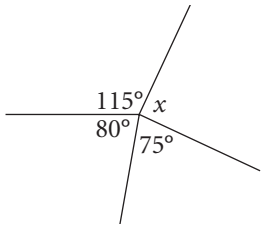

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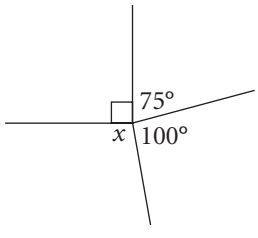
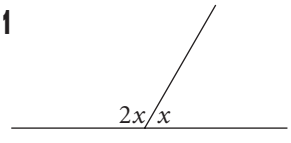


C

Find the value of the pronumeral x in each diagram.

1



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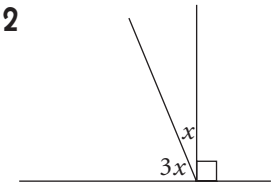

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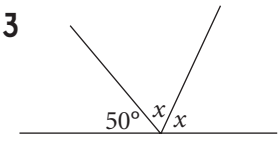

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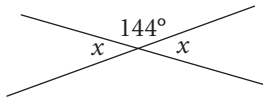

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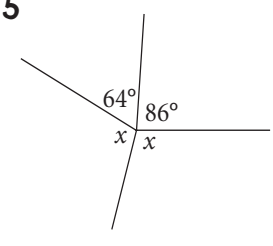

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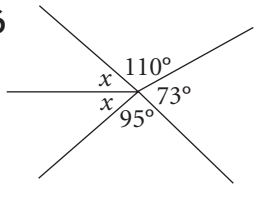

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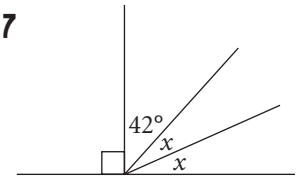

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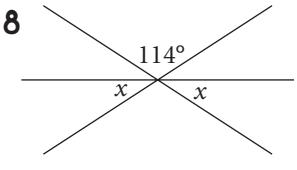

Write a sentence to show the meaning of each of the following mathematical terms.

9 revolution **10** right angle **11** acute angle

12 obtuse angle **13** adjacent angles **14** vertically opposite angles

15 straight angle **16** complementary **17** supplementary

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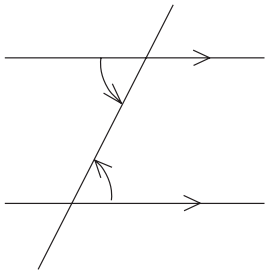
Angles and parallel lines 2.1

A Investigation

In the following diagrams, two parallel lines are cut by a third straight line (called a *transversal*).
To measure the sizes of the various angles, you may need to extend these lines to make the diagrams larger.

1



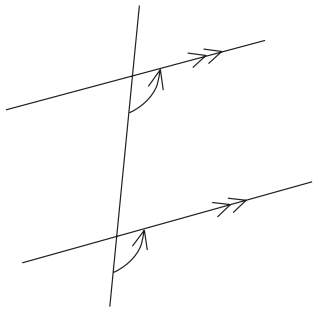
A pair of *alternate angles* have been marked.
(Alternate angles can be recognised by the 'Z-shape'.)

a Measure the size of these two angles.

b What do you notice?

Check if this is true for the pairs of alternate angles in the diagrams for questions 2 and 3.

2



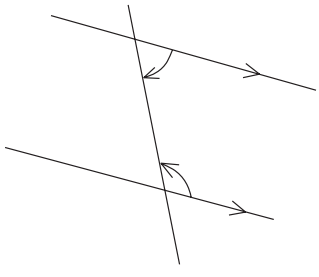
A pair of *corresponding angles* have been marked.
(Corresponding angles form an 'F-shape'.)

a Measure the size of these two angles.

b What do you notice?

Check if this is true for the pairs of corresponding angles in the diagrams for questions 1 and 3.

3



A pair of *co-interior angles* have been marked.
(Co-interior angles form a 'C-shape' or 'U-shape'.)

a Measure the size of these two angles.

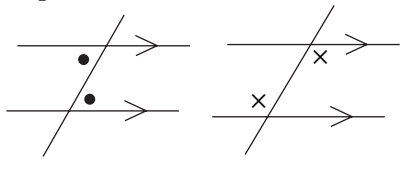
b Add them together. What is their sum?

Check if this is true for the pairs of co-interior angles in the diagrams for questions 1 and 2.

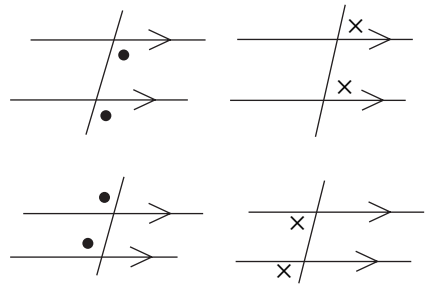
B

When two parallel lines are cut by a transversal:

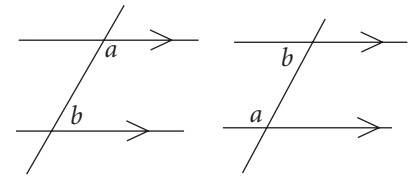
- a pair of alternate angles are equal



- a pair of corresponding angles are equal



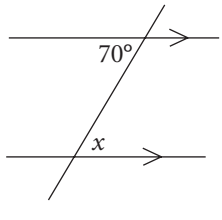
- a pair of co-interior angles add to 180°.



$a + b = 180°$

Find the size of the angle marked x in each diagram.

1



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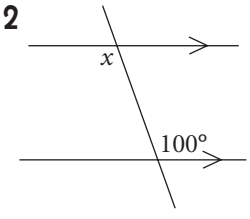

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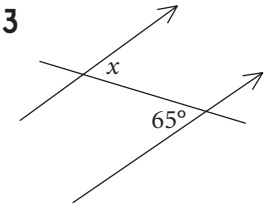

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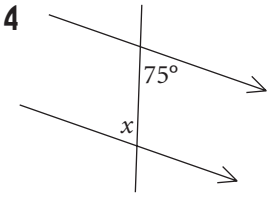

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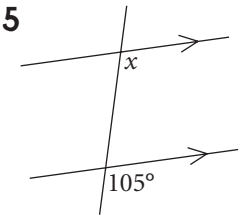

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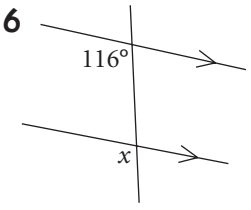

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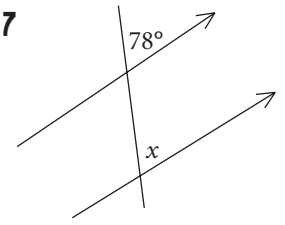

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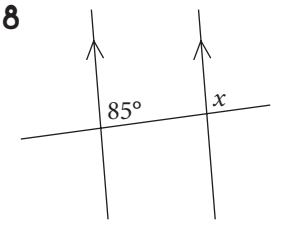

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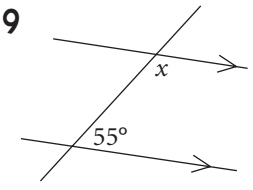

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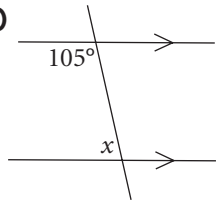

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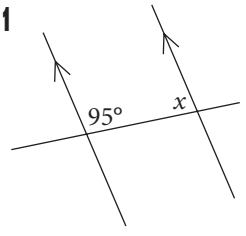

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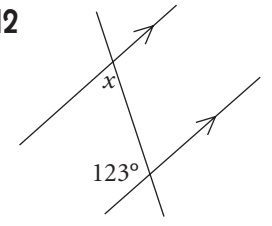

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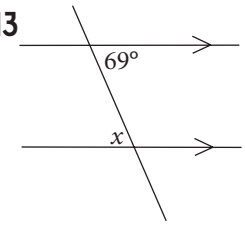

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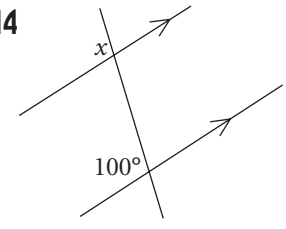

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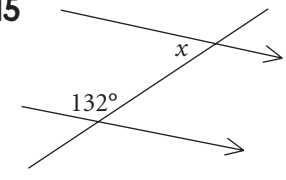

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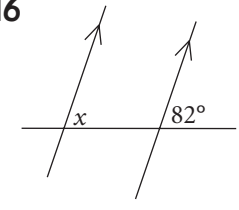

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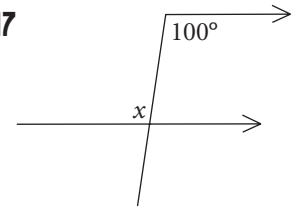

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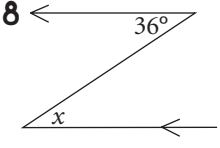

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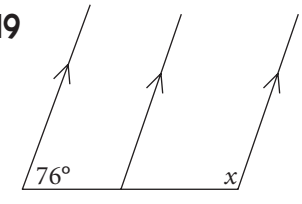

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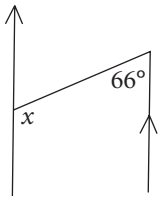


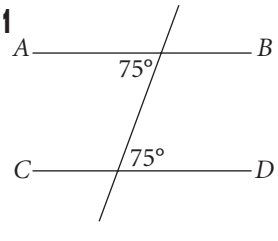
C

We can use the angle properties of parallel lines to test whether a pair of lines are parallel.
Two lines *are parallel* if

- a pair of alternate angles are equal, or
- a pair of corresponding angles are equal, or
- a pair of co-interior angles add to 180.°

In the following diagrams, test whether the lines AB and CD are parallel.Give a brief reason for each answer.

1



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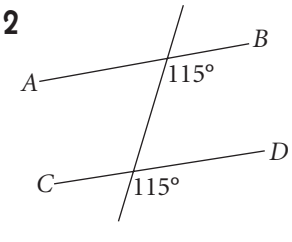

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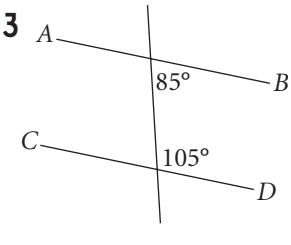

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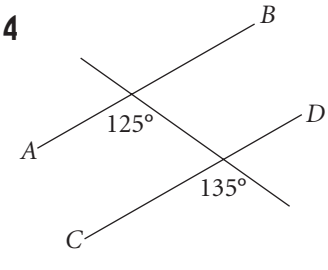

5 Use your ruler and a protractor to construct a pair of parallel lines and a transversal, given:

- a pair of corresponding angles measure 70°
- a pair of alternate angles measure 65°
- a pair of corresponding angles measure 120°
- one co-interior angle measures 70°.

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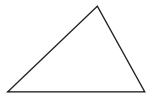
Classifying triangles 3.1

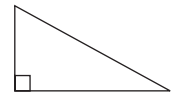
A

A *triangle* is a plane figure bounded by 3 straight sides.
It can be classified by the types of angles it contains.

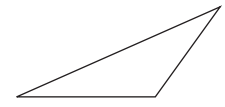
- An *acute-angled triangle* has 3 acute angles. (Each angle measures less than 90°.)



- A *right-angled triangle* has one of its angles equal to 90°.

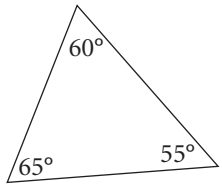


- An *obtuse-angled triangle* has one angle measuring more than 90°.



What type of triangle is each of the following? Classify using the types of angles described above.

1



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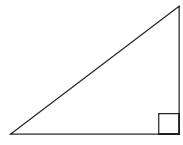

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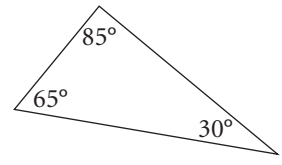

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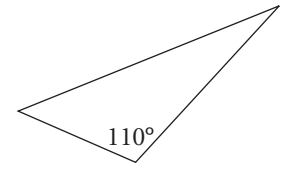

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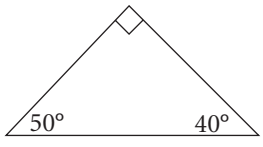

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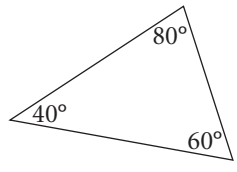

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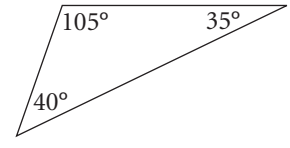

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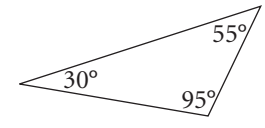
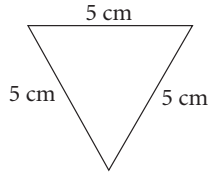


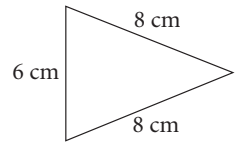
B

Triangles can also be classified according to the number of equal sides:

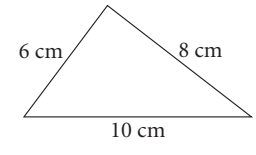
- In an *equilateral triangle*, the 3 sides are equal in length.



- In an *isosceles triangle*, 2 sides are the same length, and the third side is a different length.

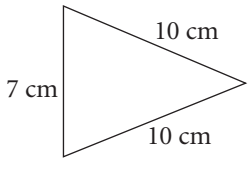


- In a *scalene triangle*, the 3 sides are different lengths.



What type of triangle is each of the following? (Equilateral, isosceles or scalene?)

1



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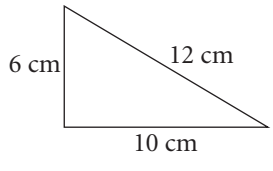

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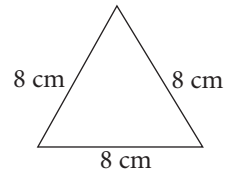

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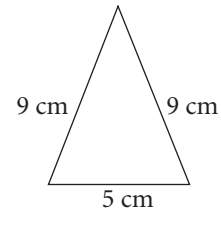

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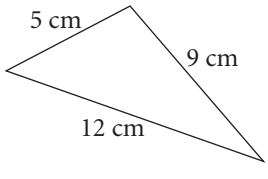

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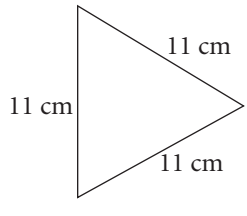

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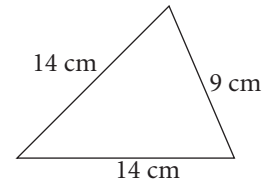

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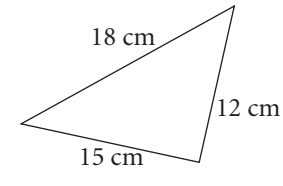

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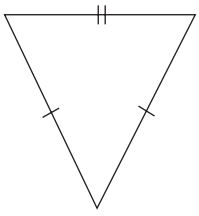

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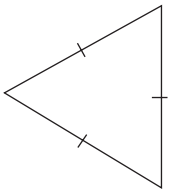

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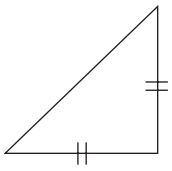

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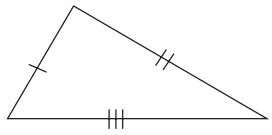
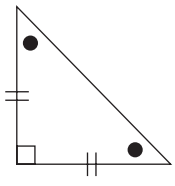


C

We often use a combination of these two methods (types of angles and lengths of sides) to classify a triangle.

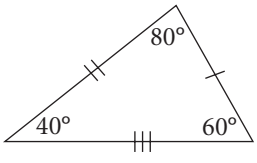
Example: This is a right-angled isosceles triangle.



Remember – if two sides of a triangle are the same length, then the angles opposite them are equal.

Classify each of these triangles, using the types of angles and the lengths of the sides.

1



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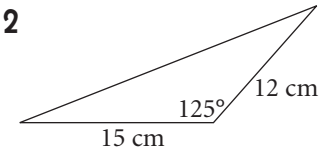

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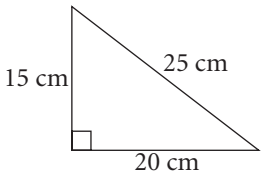

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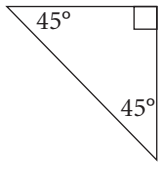

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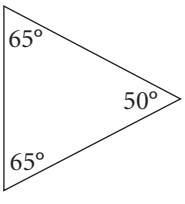

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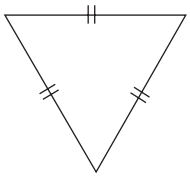

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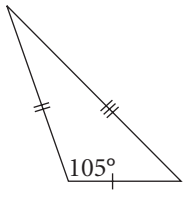

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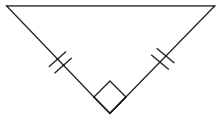

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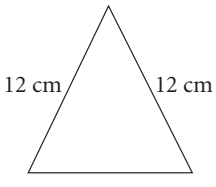

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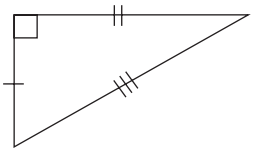

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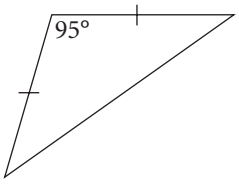

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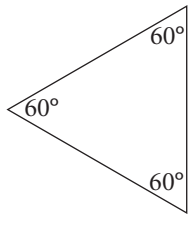

13 Draw a sketch diagram of:

- **a** an acute-angled scalene triangle
- **b** an acute-angled isosceles triangle
- **c** an obtuse-angled scalene triangle
- **d** an obtuse-angled isosceles triangle
- **e** a right-angled isosceles triangle.

14 What is the size of each angle in an equilateral triangle?

15 If a right-angled triangle is isosceles, what is the size of each of the other angles?

16 Can a triangle have two obtuse angles? Give a brief reason for your answer.

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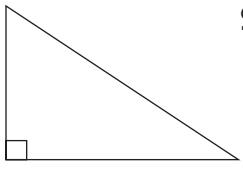
Angle properties of triangles 4.1

A

What type of triangle is each of the following? Classify using the types of angles.

1



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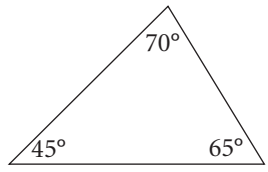

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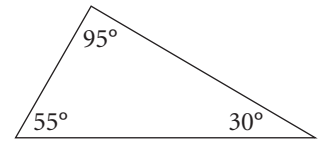

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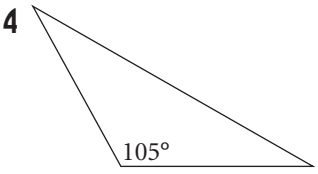

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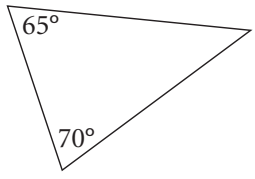

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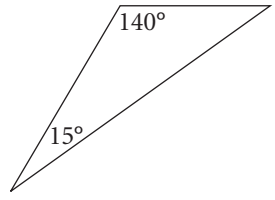

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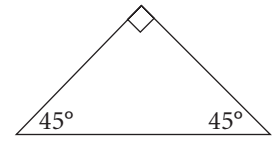

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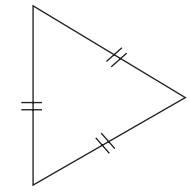
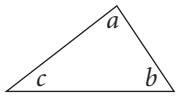
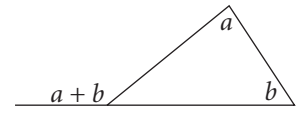


B

The sum of the angles in any triangle is 180°.
$a + b + c = 180°$

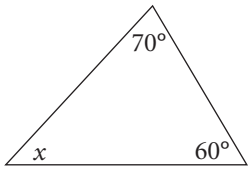


An exterior angle is equal to the sum of the two remote interior angles.



Find the size of the angle marked x in each diagram.

1



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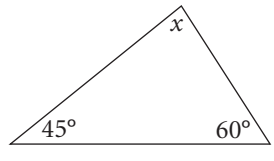

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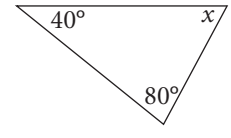

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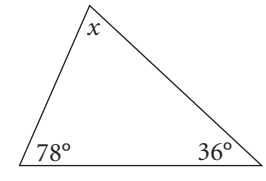

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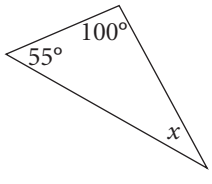

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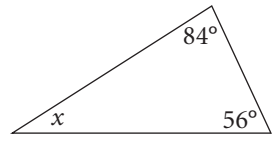

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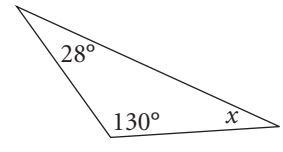

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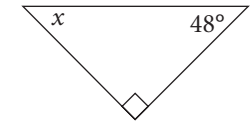

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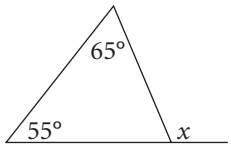

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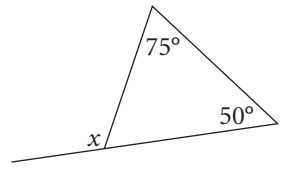

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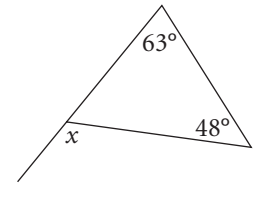

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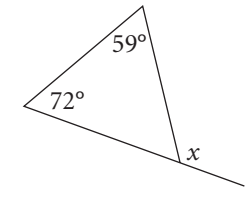

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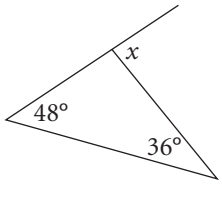

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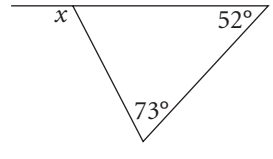

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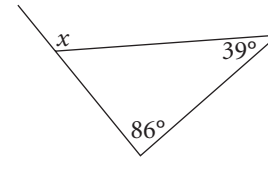

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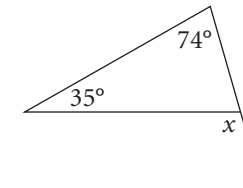

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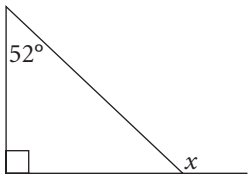

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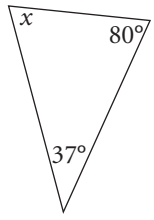

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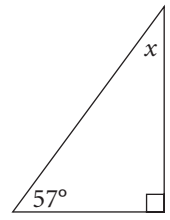

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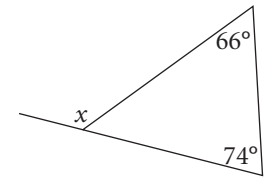

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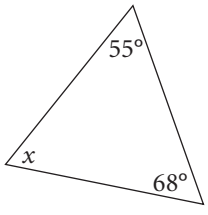

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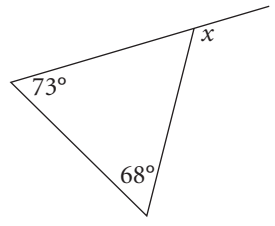

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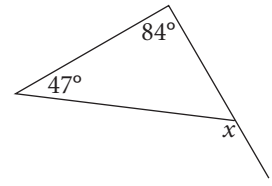

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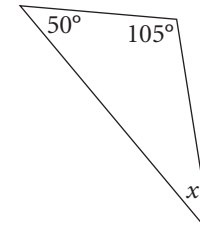

C

Find the value of each pronumeral in the following diagrams.

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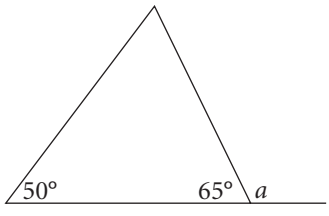

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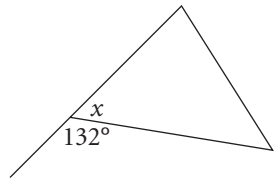

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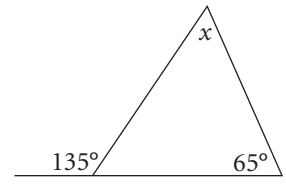

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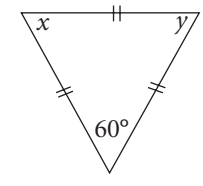

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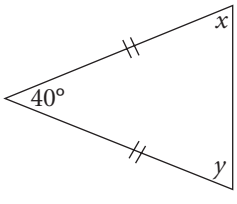

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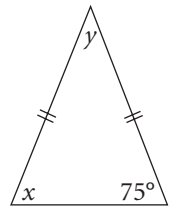

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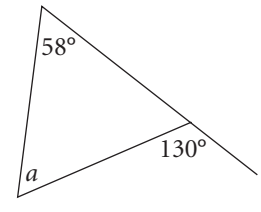

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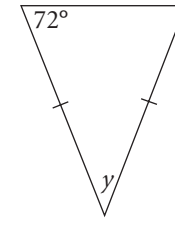

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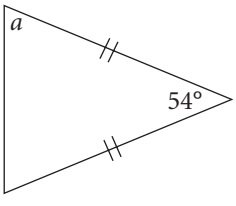

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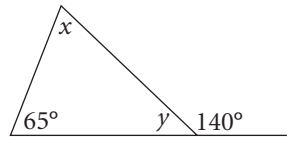

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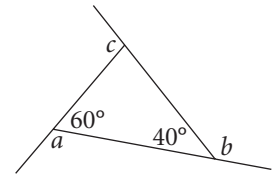

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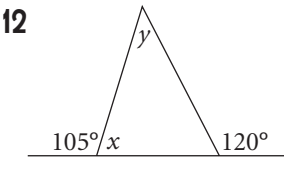

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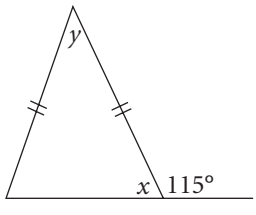

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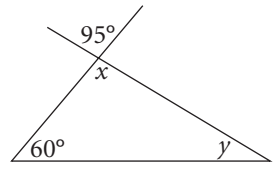

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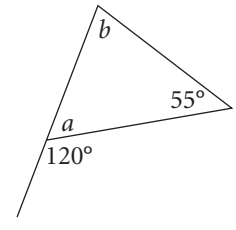

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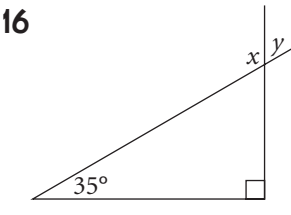

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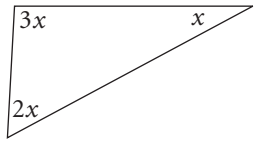

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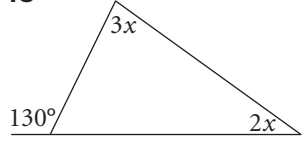

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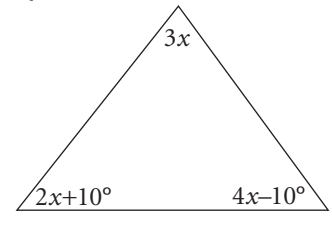

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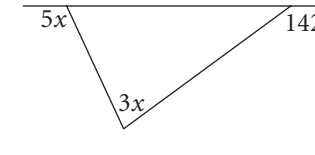

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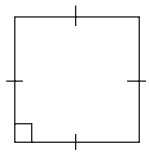
Quadrilaterals 5.1

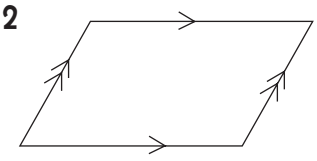
A

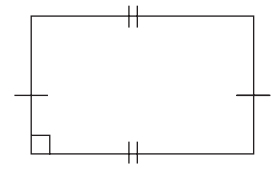
Quadrilaterals are plane figures bounded by four straight sides. They are classified according to their properties, particularly the number of pairs of parallel sides and equal sides.

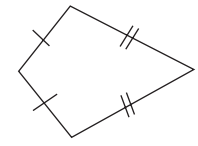
Types of quadrilaterals: parallelogram, rectangle, square, rhombus, trapezium, kite

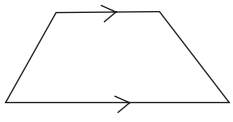
State the type of quadrilateral in each diagram.

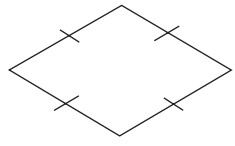
1 

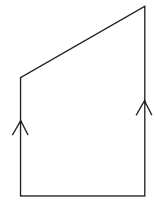
2 

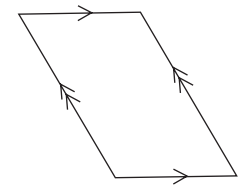
3 

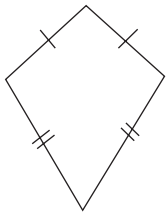
4 

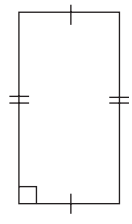
5 

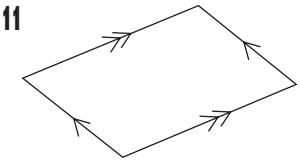
6 

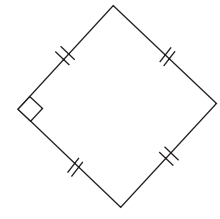
7 

8 

9 

10 

11 

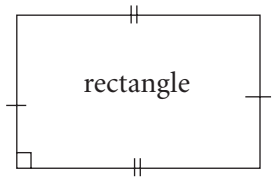
12 

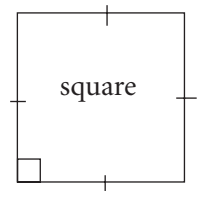
B

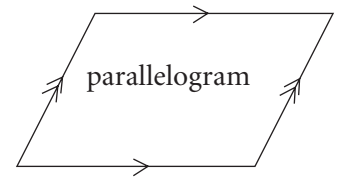
Many quadrilaterals have symmetry properties.

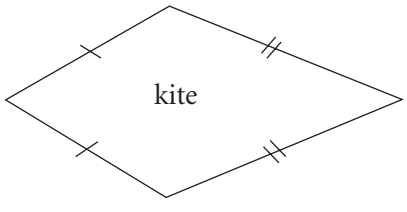
A figure has *line symmetry* if it can be folded along a line, or reflected in a mirror, so that the two halves match exactly. The line of the fold is called an *axis of symmetry*.

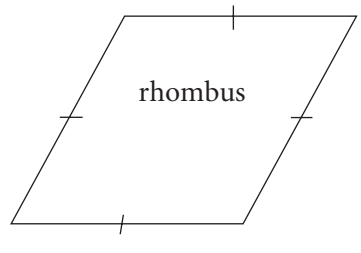
Investigate the line symmetry of these quadrilaterals. You may need to copy each figure onto tracing paper and cut out your copy. Use dotted lines to show each axis of symmetry on the quadrilaterals below, and state the number of axes of symmetry for each shape.

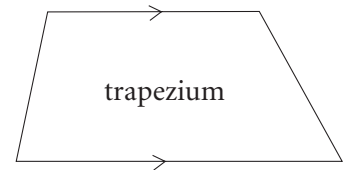
1 

2 

3 

4 

5 

6 

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A plane shape has *rotational symmetry* if, when you spin it through one revolution, it matches its original position at least twice. The numbers of times it matches its original position in one revoluion determines the *order of symmetry*.

Investigate the rotational symmetry of the six types of quadrilaterals using the diagrams in questions 1 to 6. Then state the order of rotational symmetry of:

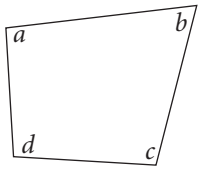
7 a rectangle **8** a square **9** a parallelogram

10 a kite **11** a rhombus **12** a trapezium

C

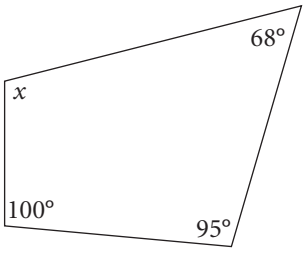
The four angles of any quadrilateral add to 360°.

$a + b + c + d = 360°$



Find the size of the angle marked x in each diagram.

1



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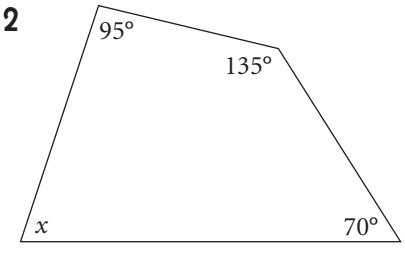

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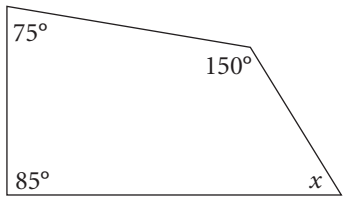

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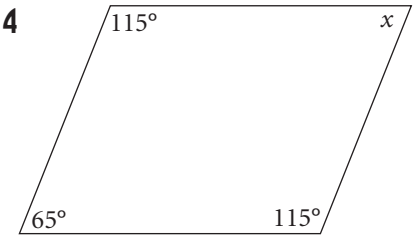

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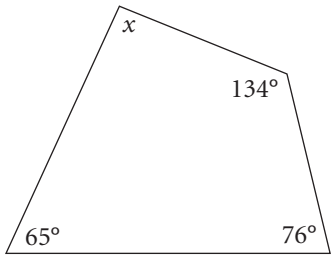

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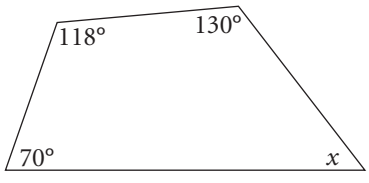

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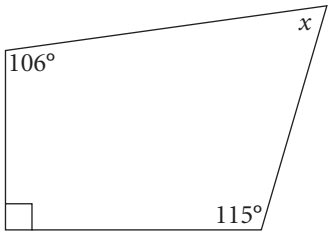

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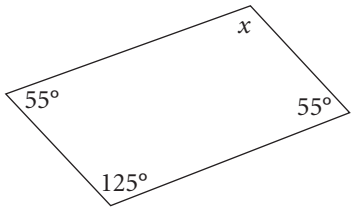

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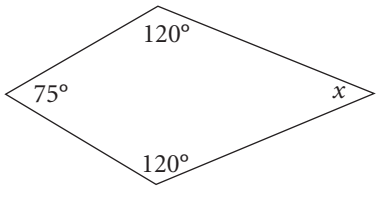

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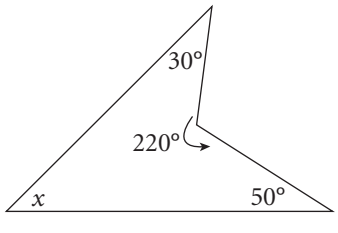

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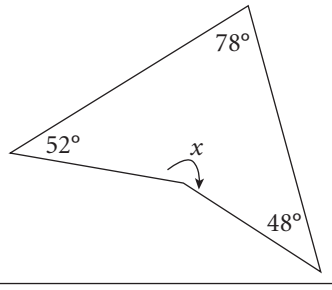

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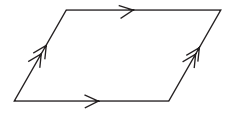
Properties of a parallelogram 6.1

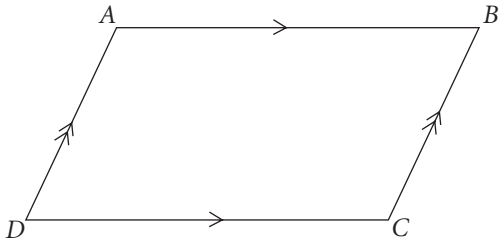
A

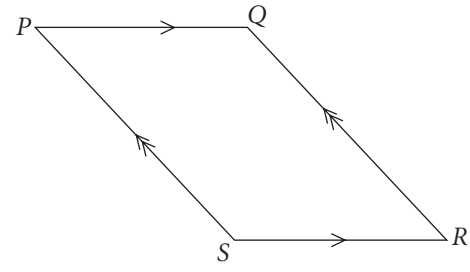
A *parallelogram* is a quadrilateral in which both pairs of opposite sides are parallel.



Investigation

ABCD and *PQRS* are two parallelograms.





1 Use a ruler to measure the length of each side of *ABCD* to the nearest millimetre.

2 Measure the length of each side of *PQRS* to the nearest millimetre.

3 Are the opposite sides, in each parallelogram, the same length?

4 Use a protractor to measure the angles of each parallelogram.

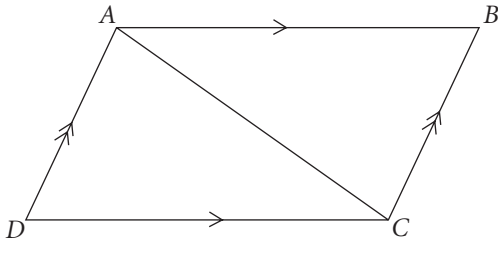
5 In each parallelogram, are the opposite angles the same size? (i.e. Does $\angle A = \angle C$ and $\angle B = \angle D$?)

6 On the diagram *ABCD*, draw the diagonals *AC* and *BD*. Measure the length of each diagonal to the nearest millimetre.

7 Does $AC = BD$?

8 Do the diagonals *AC* and *BD* cut each other in half?

9 On the diagram *PQRS*, draw the diagonals *PR* and *QS*. Do the diagonals *PR* and *QS* cut each other in half?



This diagram is a copy of the parallelogram *ABCD*.
The diagonal *AC* has been drawn.

10 Using a protractor, measure the size of $\angle BAC$ and $\angle DAC$. Are they the same size?

11 Measure the size of $\angle BCA$ and $\angle DCA$. Are they the same size?

12 Does the diagonal *AC* bisect the angles of the parallelogram? (Does it cut $\angle BAD$ and $\angle BCD$ in half?)

13 Complete this summary using the wordbank to fill in the missing words.

A *parallelogram* has the following properties:

- There are _____ pairs of parallel sides.
- The _____ sides are the same length.
- The opposite angles are the _____ size.
- The diagonals are _____ lengths.
- The diagonals _____ each other (i.e. cut each other in half).

Word bank: *bisect, different, opposite, same, two.*
(Each word may be used once only.)

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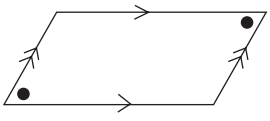
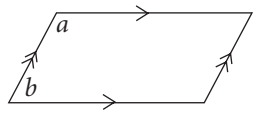
B

Angle properties of a parallelogram:

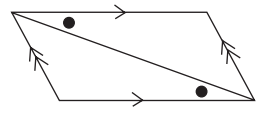
- Opposite angles are equal.



- Co-interior angles add to give 180°. $a + b = 180°$



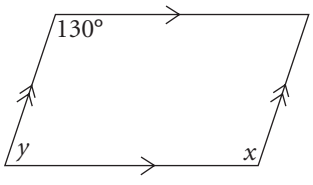
- Alternate angles formed by a diagonal are the same size.



- The four angles add to 360°.

Find the size of each angle marked with a pronumeral.

1



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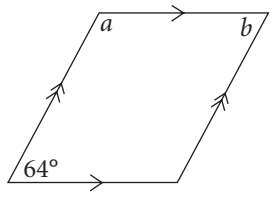

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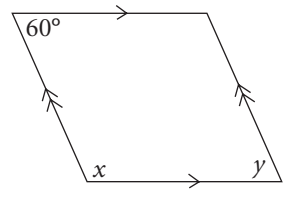

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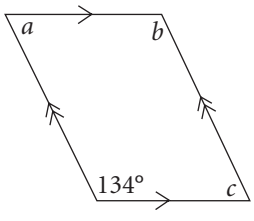

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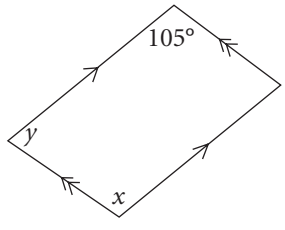

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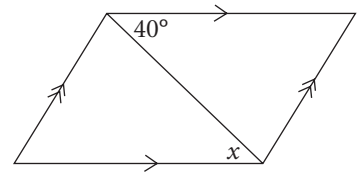

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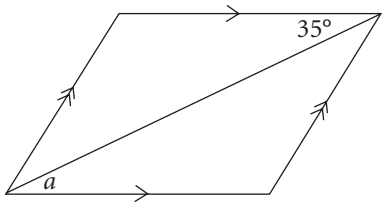

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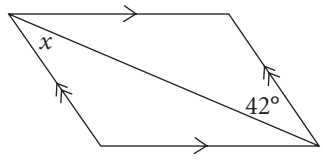

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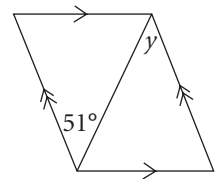
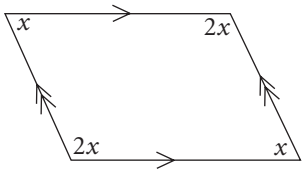


C

Find the value of each pronumeral in the following diagrams.

1



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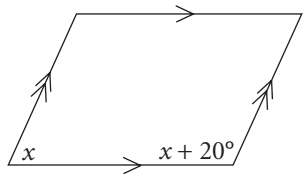

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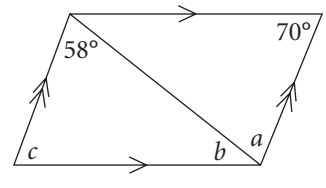

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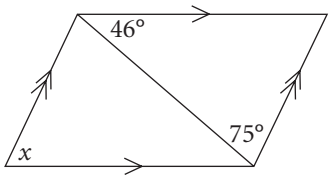

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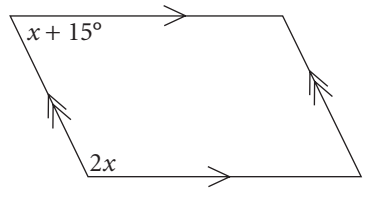

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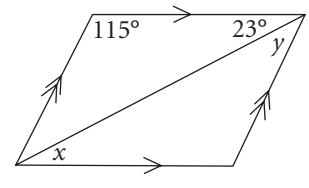

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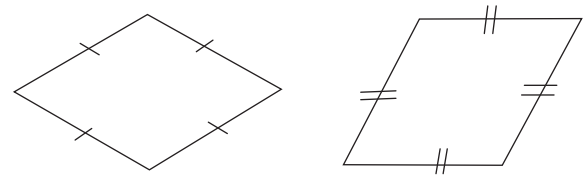
NOTE: DIAGRAMS NOT TO SCALE

Properties of a rhombus 7.1

A

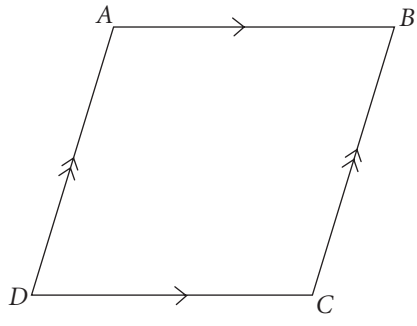
A *rhombus* (or diamond) is a quadrilateral with four equal sides.

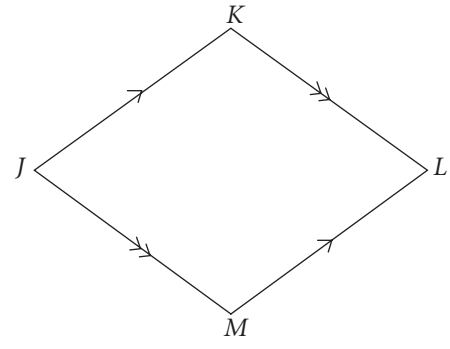
A rhombus can also be defined as a parallelogram with four equal sides.



Investigation

ABCD and *JKLM* are two rhombuses.





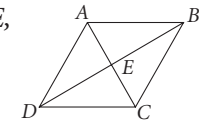
1 Use your ruler to measure the length of each side of *ABCD* to the nearest millimetre. Confirm that the four sides *AB*, *BC*, *CD* and *DA* are the same length.

2 Measure the length of each side of *JKLM* to the nearest millimetre. Confirm that the four sides *JK*, *KL*, *LM* and *MJ* are the same length.

3 Use a protractor to measure the angles of each figure.

4 In each rhombus, are the opposite angles the same size? (ie. does $\angle A = \angle C$, $\angle B = \angle D$ and $\angle J = \angle L$, $\angle K = \angle M$?)

5 On the diagram *ABCD* above, draw the diagonals *AC* and *BD*. Label their point of intersection *E*, as shown on the right.



6 Measure the length of each diagonal to the nearest millimetre. Does $AC = BD$?

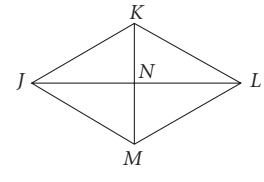
7 Measure the lengths of *AE* and *EC* to the nearest millimetre. Does $AE = EC$?

8 Measure the lengths of *BE* and *ED* to the nearest millimetre. Do the diagonals *AC* and *BD* cut each other in half?

9 Measure the size of $\angle AEB$.

10 Measure the size of $\angle ABE$ and $\angle CBE$. Does the line *BD* cut $\angle ABC$ in half?

11 On the diagram *JKLM* above, draw the diagonals *JL* and *KM*. Label their point of intersection *N*, as shown on the right.



12 Do the diagonals *JL* and *KM* cut each other in half? (ie. does $JN = NL$ and $KN = NM$?)

13 Measure the size of $\angle KNL$.

14 Measure the size of $\angle JMN$ and $\angle LMN$. Does the line *KM* cut $\angle JML$ in half?

15 Measure the size of $\angle KLN$ and $\angle MLN$. Does the diagonal *JL* bisect $\angle KLM$ (cut it in half)?

16 Complete this summary using the wordbank to fill in the missing words.

Wordbank: *all, angles, diagonals, lengths, parallel, right, same.* (Each word may be used once only.)

A *rhombus* has the following properties:

- _____ sides are the same length.
- There are two pairs of _____ sides.
- Opposite angles are the _____ size.
- The diagonals are different _____.
- The diagonals meet at _____ angles.
- The _____ bisect each other (cut each other in half).
- The diagonals bisect the _____of the rhombus.

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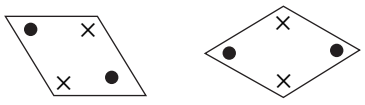
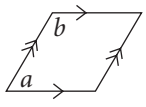
B

Angle properties of rhombus:

- Opposite angles are equal.

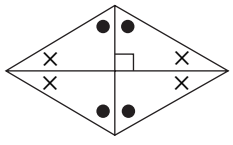


- Co-interior angles add to give 180°.



$a + b = 180°$

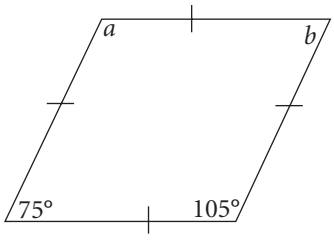
- The diagonals meet at right-angles.
- The diagonals bisect the angles of the rhombus.



- The four angles of a rhombus add to 360°.

Find the size of each angle marked with a pronumeral in the following diagrams.

1



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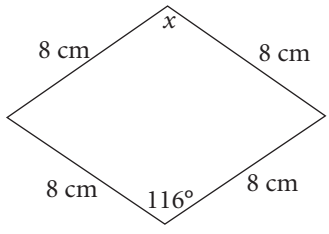

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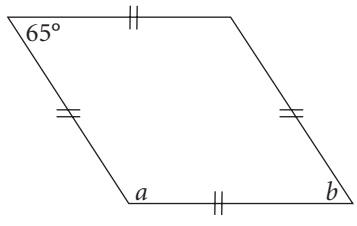

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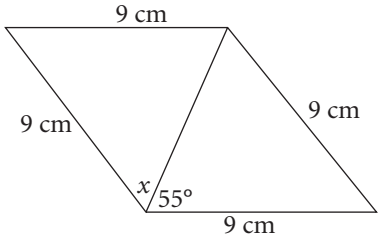

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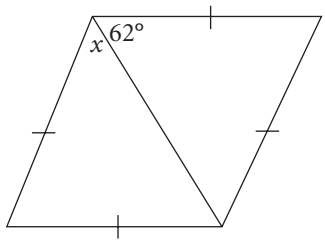

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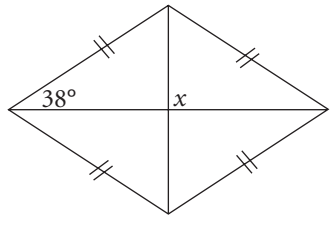

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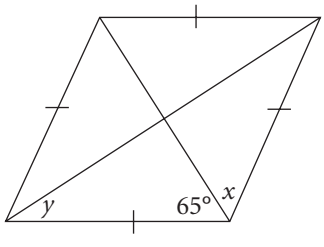

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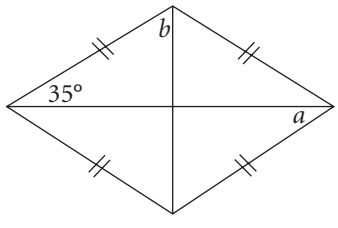

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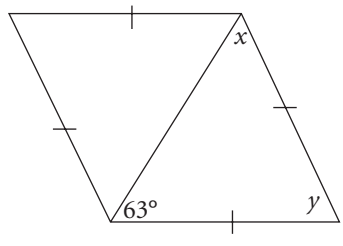
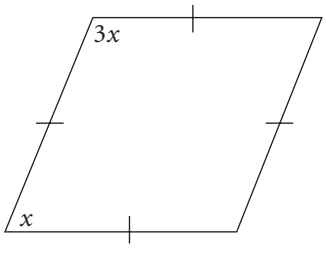


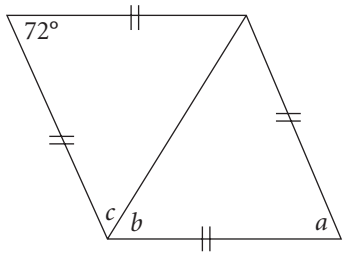
C

Find the value of each pronumeral in the following diagrams.

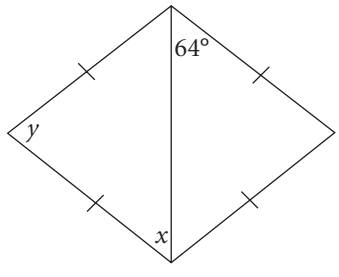
1



2



3



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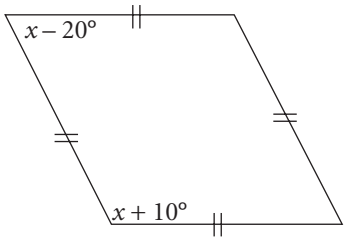

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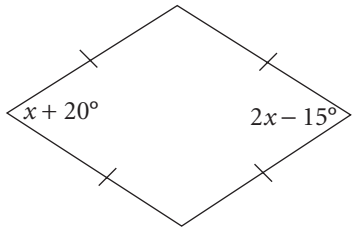

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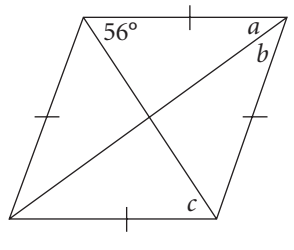

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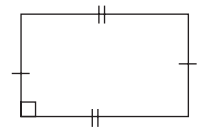
Properties of a rectangle 8.1

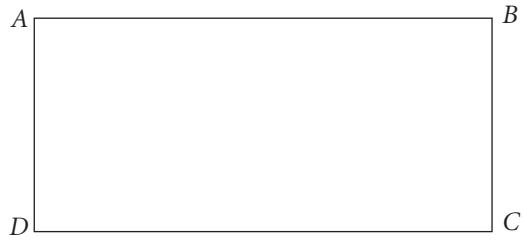
A

A *rectangle* is a parallelogram in which all angles measure 90°.



Investigation

ABCD is a rectangle.

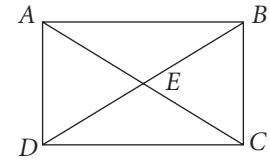


1 Use a ruler and a protractor to measure the length of each side and the size of each angle.

2 Are the opposite sides equal in length? (Does $AB = DC$ and $AD = BC$?)

3 Do all angles measure 90°?

4 Draw the two diagonals *AC* and *BD* on the diagram above and label their point of intersection *E*, as shown on the right.



5 Measure the lengths of the diagonals *AC* and *BD* to the nearest millimetre. Does $AC = BD$?

6 Measure the lengths of *AE*, *EC*, *BE* and *ED*. Does $AE = EC$ and $BE = ED$?

7 Complete the sentence: 'The diagonals of a rectangle cut each other _____'

8 Do the diagonals meet at right angles? (Does $\angle AEB$ measure 90°?)

9 Complete this summary using the wordbank to fill in the missing words. (Remember: a rectangle has all the properties of a parallelogram.)

A *rectangle* has the following properties:

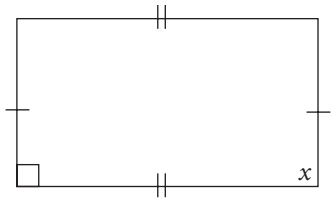
- _____ sides are the same length.
- Opposite sides are _____.
- All angles are _____ angles (measure 90°).
- The diagonals are the same _____.
- The _____ bisect each other (cut each other in half).

Wordbank: *diagonals, length, opposite, parallel, right.* (Each word may be used once only.)

B

Find the size of each angle marked with a pronumeral in the following diagrams.

1



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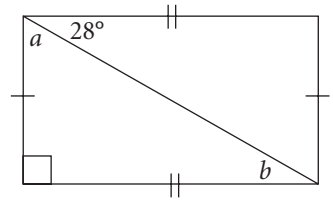
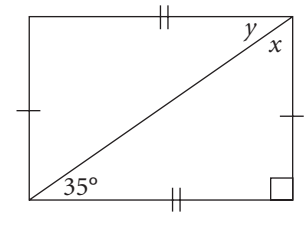
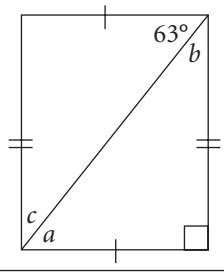


3



4



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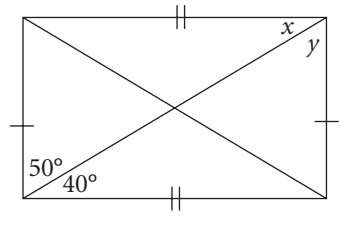

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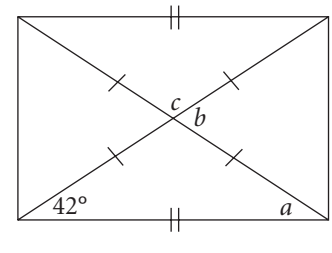

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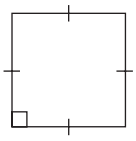
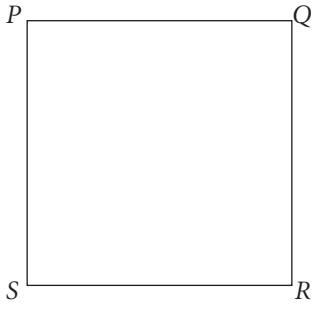
Properties of a square 9.1

A

A *square* is a rectangle with four equal sides.
A square can also be defined as a parallelogram with four equal sides and four right angles.

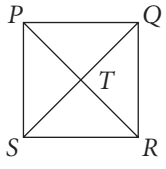


PQRS is a square.



1 Use a ruler to measure the length of each side to the nearest millimetre. Are the four sides the same length?

2 Draw the diagonals *PR* and *QS* on the diagram above. Label their point of intersection *T*, as shown on the right.



3 Measure the lengths of the diagonals *PR* and *QS* to the nearest millimetre. Are the diagonals the same length?

4 Measure the lengths of *PT*, *TR*, *ST* and *TQ*. Does $PT = TR$ and $ST = TQ$?

5 Complete the sentence: 'The diagonals of a square cut each other _____'

6 Measure the size of $\angle PTQ$.

7 Complete the sentence: 'The diagonals of a square meet at _____'

8 Measure the size of $\angle PST$ and $\angle RST$. Does $\angle PST = \angle RST$?

9 Measure the size of $\angle SRT$ and $\angle QRT$. Are they equal?

10 Do the diagonals bisect the angles of the square?

11 Complete this summary using the wordbank to fill in the missing words.

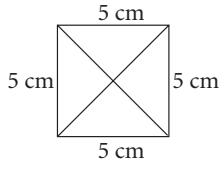
A *square* has the following properties:

- All four sides are the _____ length.
- The opposite sides are _____.
- All four angles are _____ angles.
- The diagonals are the same _____.
- The diagonals _____ each other (cut each other in half).
- The _____ meet at right angles.
- The diagonals bisect the _____ of the square.

Wordbank: *angles, bisect, diagonals, length, parallel, right, same.*
(Each word may be used once only.)

B

Draw a square with sides 5 cm long on a piece of grid paper.
Draw the diagonals of the square, as shown on the right.
Cut out the square and cut along the diagonals to give four identical triangular pieces.



Rearrange these four pieces to form

1 a rectangle **2** a parallelogram **3** a trapezium **4** a triangle.

You must use *all* four pieces each time.

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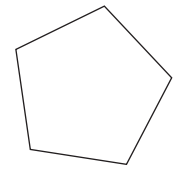
Polygons 10.1

A

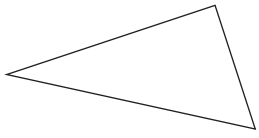
A *polygon* is a plane shape bounded by straight sides, which do not cross over.
Polygons are named according to the number of sides they have:

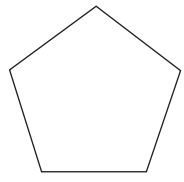
- 3 sides – a triangle
- 4 sides – a quadrilateral
- 5 sides – a pentagon
- 6 sides – a hexagon
- 7 sides – a heptagon
- 8 sides – an octagon
- 9 sides – a nonagon
- 10 sides – a decagon
- 12 sides – a dodecagon

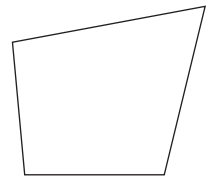


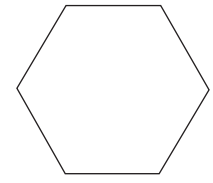
A *regular polygon* has equal sides and equal angles.

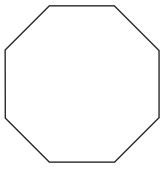
Name each of the following polygons.

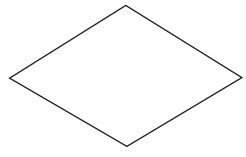
1 

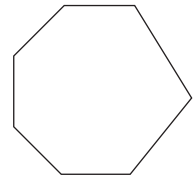
2 

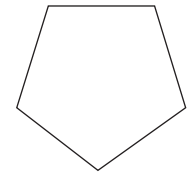
3 

4 

5 

6 

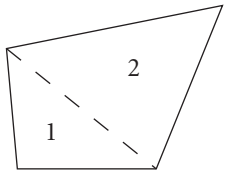
7 

8 

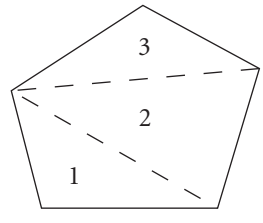
B

The sum of the angles of a polygon

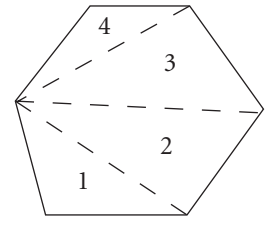
In the following diagrams, all possible diagonals have been drawn *from one vertex (corner) only*, dividing each polygon into triangles. Note that the angles of the triangles form the angles of the polygon.



quadrilateral



pentagon



hexagon

1 Draw similar diagrams for a heptagon (7 sides) and an octagon (8 sides).

2 Count the number of triangles formed in each diagram. Complete the following table, calculating the sum of the angles of each polygon.

Polygon	Number of sides	Number of triangles	Angle sum
triangle	3	1	180°
quadrilateral	4	2	$2 \times 180° = 360°$
pentagon	5	3	$3 \times 180° = 540°$
hexagon	6	4	$4 \times 180° =$
heptagon	7		
octagon	8		
nonagon	9		
decagon	10		

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3 How do you find the number of triangles when you know the number of sides (without drawing a diagram)?

4 Write a rule to find the sum of the angles of a polygon with n sides.

In a *convex polygon*, each interior angle measures less than 180°.
The sum of the angles of a convex polygon with n sides add up to $(2n - 4)$ right angles *or* $(n - 2) \times 180°$.

Example: The sum of the angles of a dodecagon (12 sides) is:
$(12 - 2) \times 180° = 10 \times 180° = 1800°$

5 Calculate the sum of the angles of a polygon with:

a 9 sides **b** 15 sides **c** 16 sides **d** 11 sides
e 20 sides **f** 14 sides **g** 18 sides **h** 25 sides

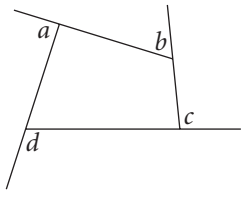
6 A *regular polygon* has equal sides and equal angles.
Find the size of each angle in a:

a regular pentagon **b** regular hexagon **c** regular octagon
d regular decagon **e** a regular nonagon **f** a regular dodecagon (12 sides)

C Investigation

1 Construct any quadrilateral on a piece of paper. Form the exterior angles by producing each side in rotational order, as shown. Cut out the four exterior angles (a, b, c and d) and arrange them with their vertices next to each other.

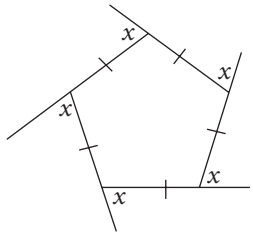
What is the sum $a + b + c + d$?



2 **a** What is the size of each angle of a regular pentagon?

b The diagram shows a regular pentagon. The exterior angles have been drawn. What is the value of x?

c Find the sum of these exterior angles.

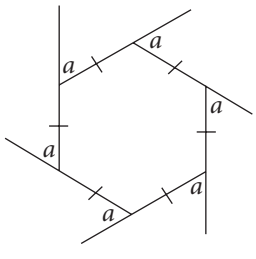


3 The diagram shows a regular hexagon. The exterior angles have been drawn.

a What is the size of each angle of a regular hexagon?

b What is the size of each exterior angle marked a?

c What is the sum of the exterior angles of a regular hexagon?



4 What is the size of each exterior angle in

a a regular octagon?

b a regular nonagon?

c a regular decagon?

d a regular dodecagon (12 sides)?

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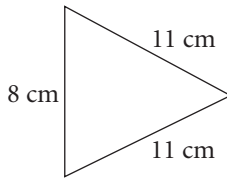
Review — properties of plane shapes 11.1

A What type of triangle (equilateral, isosceles or scalene) is shown in each diagram?

1



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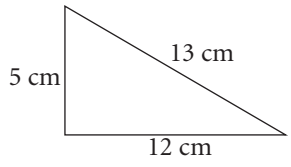

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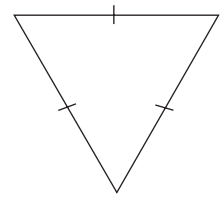

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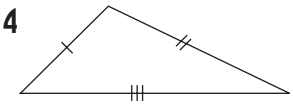
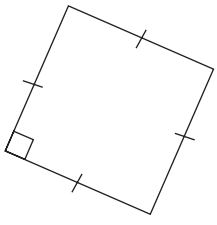


What type of quadrilateral is shown in each diagram?

5



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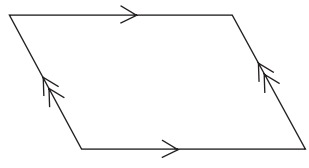

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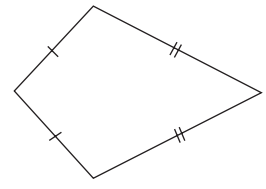

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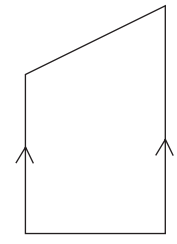
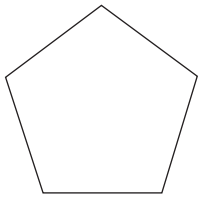


What type of polygon is shown in each diagram?

9



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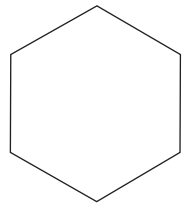

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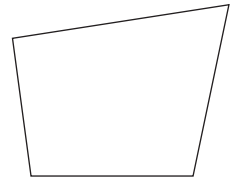

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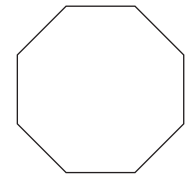
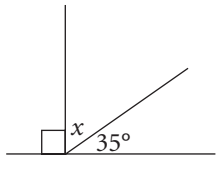


B Find the size of the angle marked x in each diagram.

1



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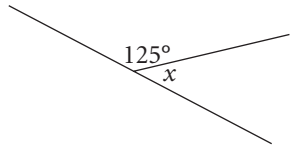

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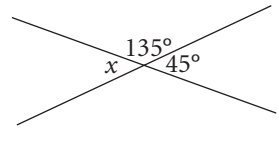

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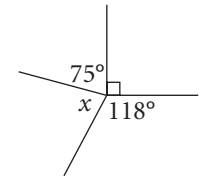

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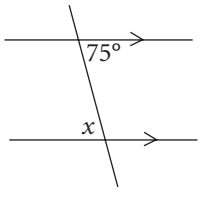

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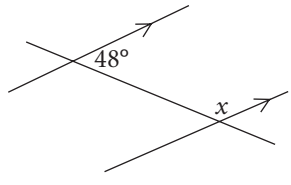

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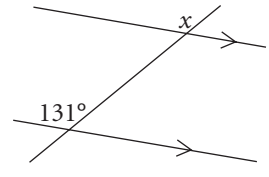

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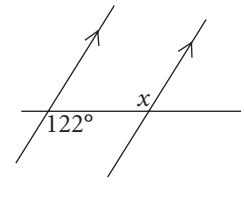

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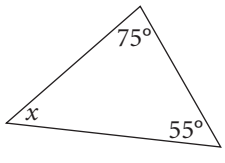

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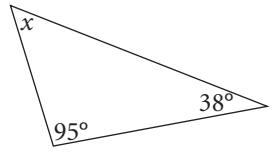

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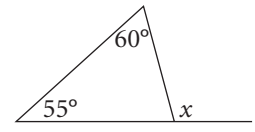

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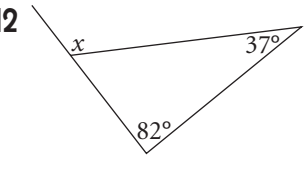

Draw a neat sketch of these quadrilaterals. Use dotted lines to show all axes of symmetry on your diagrams.

13 rectangle **14** square **15** rhombus (diamond) **16** kite

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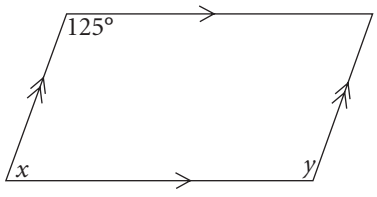
Find the size of each angle marked with a pronumeral.

17



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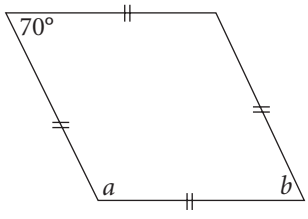

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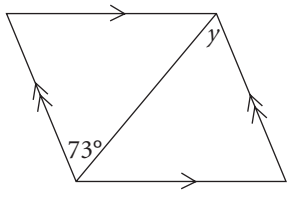

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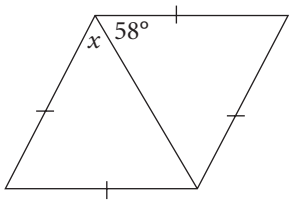

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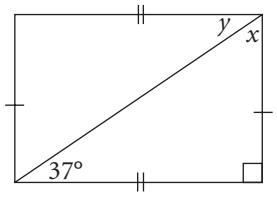

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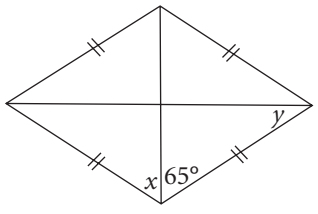


Calculate the sum of the angles of each of the following polygons.

23 hexagon

24 decagon

25 dodecagon (12 sides)

Find the size of each angle in:

26 a regular pentagon

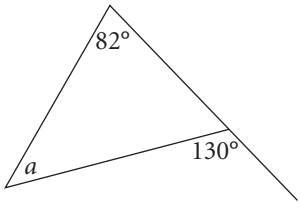
27 a regular octagon

28 a regular decagon

C

Find the value of each pronumeral in the following diagrams.

1



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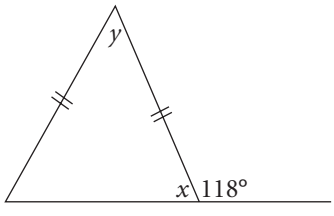

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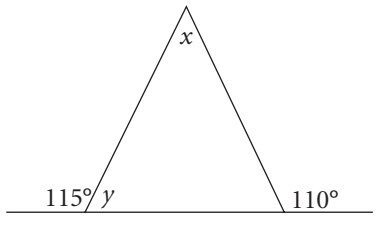

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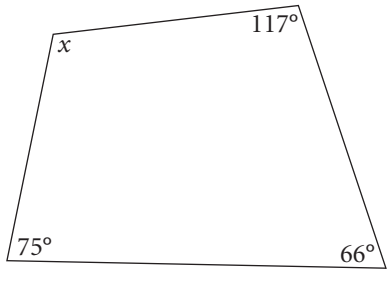

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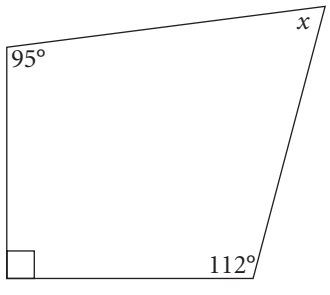

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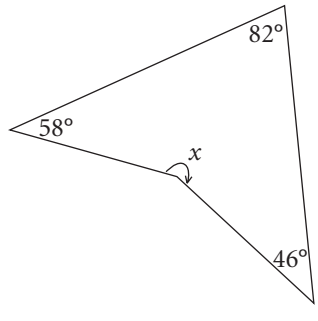
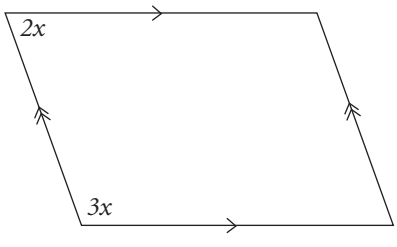
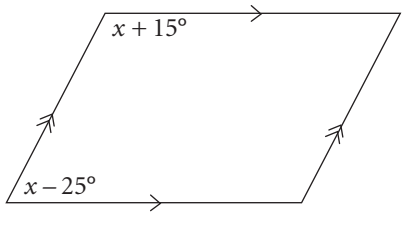


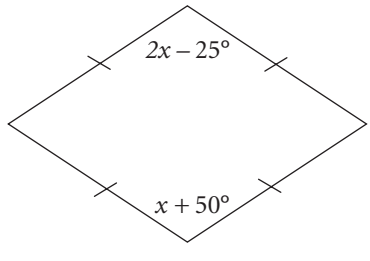
7



8



9



Write a brief paragraph to describe each of the following quadrilaterals.

10 a parallelogram

11 a rhombus

12 a rectangle

13 a square

14 a trapezium

15 a kite

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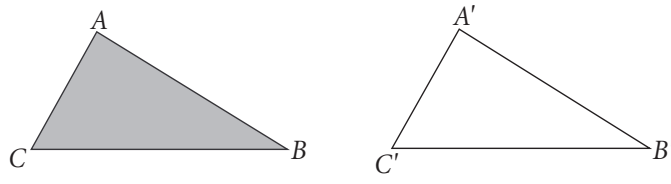
NOTE: DIAGRAMS NOT TO SCALE

Translation 12.1

Moving a plane shape to a new position by sliding is called a *translation*. In a translation, each point is moved the *same distance* in the *same direction*.

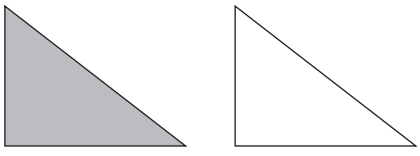


A' is the image of the original point A. An arrow (or vector) is sometimes used to indicate the direction and distance of the translation. The translated image is the same size and shape as the original figure.

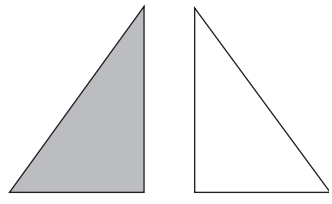
A

Are the following transformations translations?

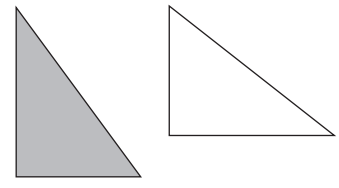
1



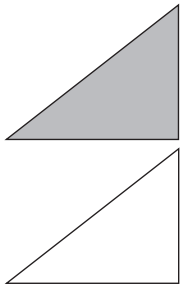
2



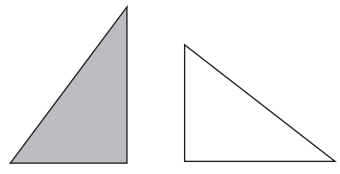
3



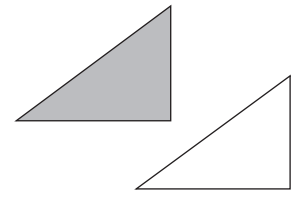
4



5

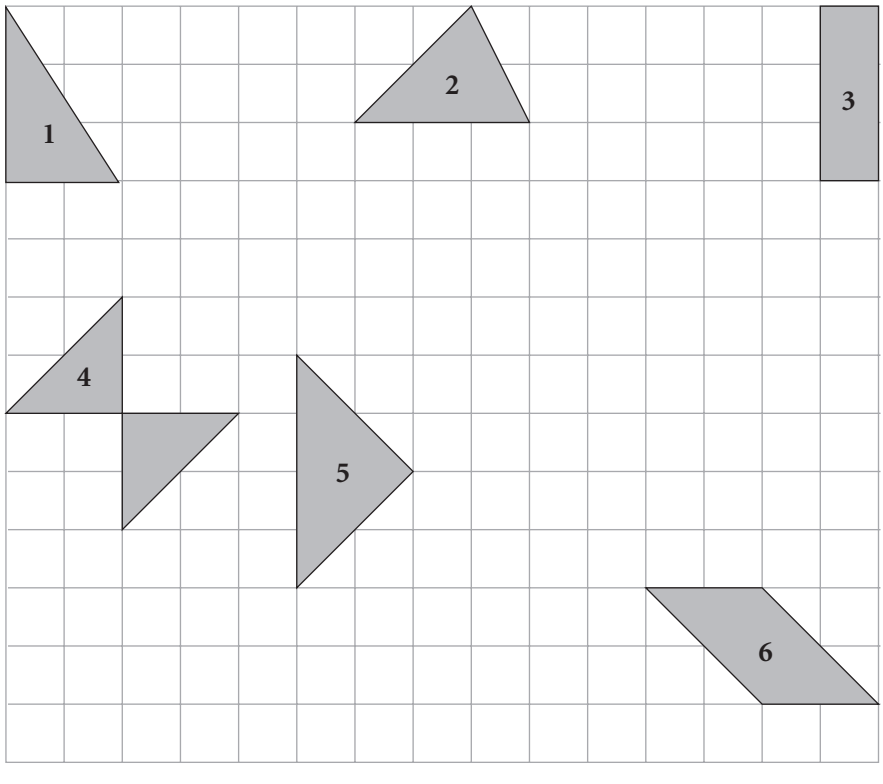


6



B

Translate each figure as instructed at right. Move each vertex first, then join the points to give the image of the original figure. (Each square of the grid has a length of 1 unit.)



1 translate 3 units to the right

2 translate 2 units down

3 translate 4 units to the left

4 translate 4 units down

5 translate 2 units to the right

6 translate 3 units up

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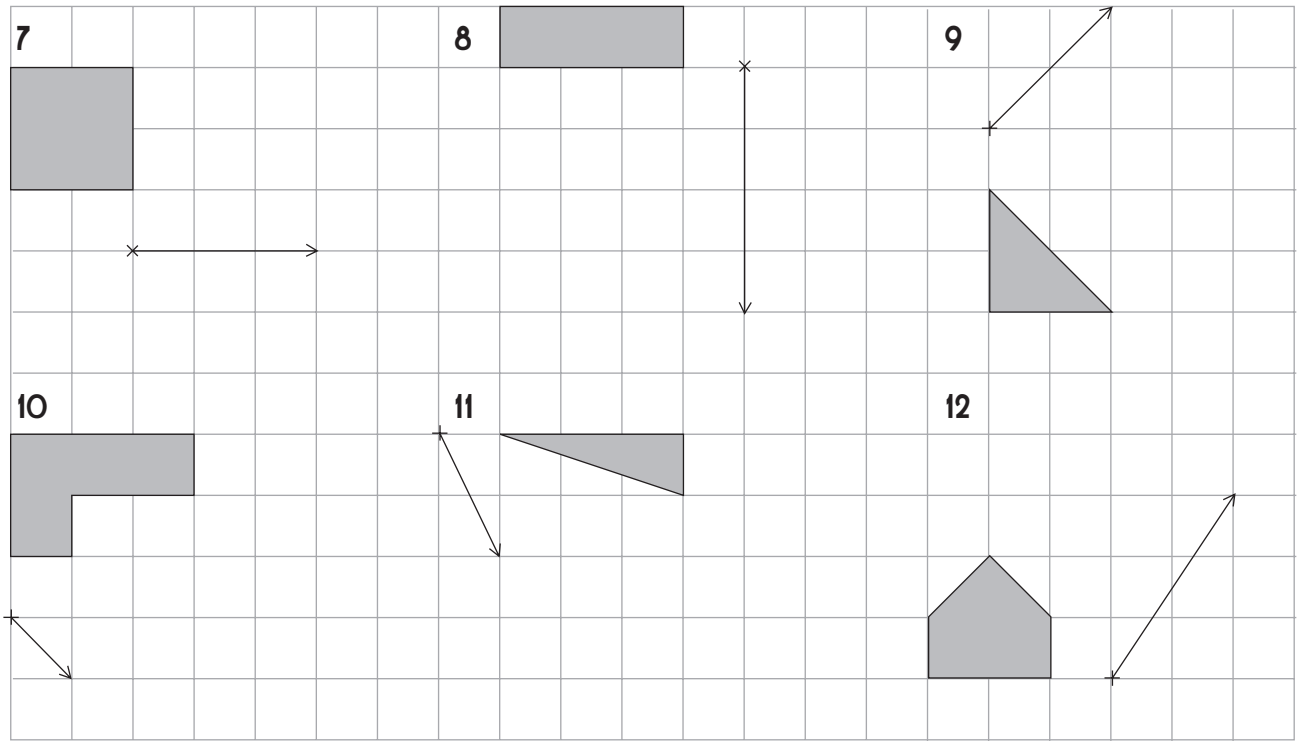
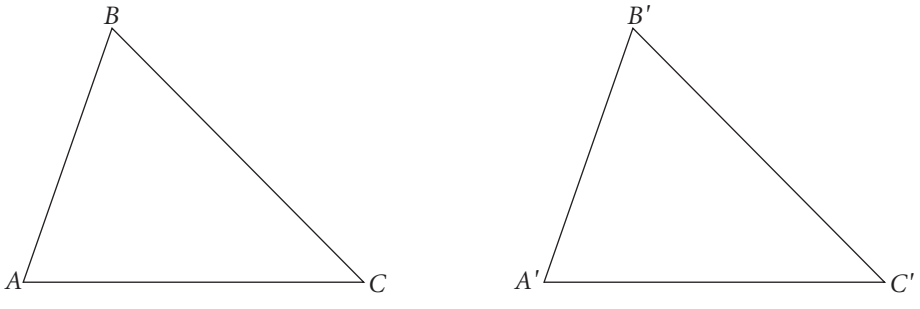
Translate these shapes according to the distance and direction indicated by the vector (arrow).



C The triangle ABC has been translated to a new position $A'B'C'$.



1 Measure the distance AA' to the nearest millimetre.

2 Measure the distances BB' and CC'. Did all points move the same distance?

3 Describe this translation (distance and direction).

4 Measure the sides of each triangle. Are matching sides the same length?

5 Measure the size of $\angle ABC$ and $\angle A'B'C'$. Are they equal?

6 Are the vertices A, B, C and A', B', C' in the same rotational order (clockwise or anticlockwise)?

7 Complete this summary using the wordbank to fill in the missing words.

Under a translation,

- all points are moved the same _____ in the same direction.
- a line and its _____ are the same length.
- an angle and its image are the same _____ .
- a _____ and its image have the same area.
- a figure and its image are _____.
- the rotational order of the _____ of the figure is unchanged.

Wordbank: *congruent, distance, figure, image, size, vertices.*
(Each word may be used once only.)

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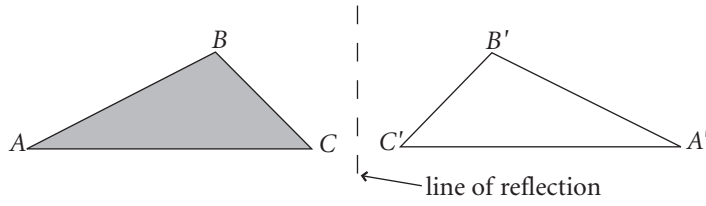
Reflection 13.1

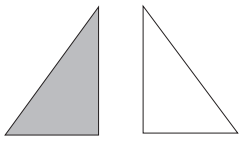
Moving a plane shape to a new position by turning it over (or flipping it over) is called a *reflection*. In a reflection, every point is reflected in a line, as occurs in a mirror. The line is called the *line of reflection*.

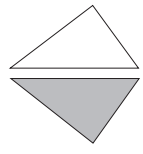


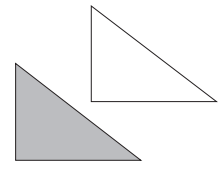
A′ is the image of point A. The line of reflection is halfway between a point and its image. Under reflection, the image is the same shape and size as the original figure, but is turned over.

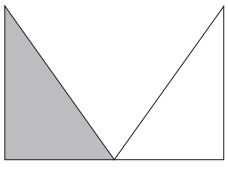
A

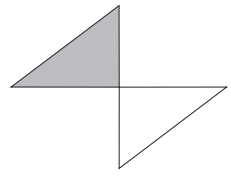
Are the following transformations reflections?

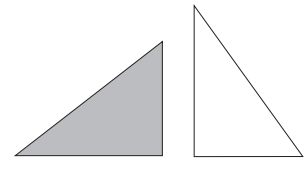
1 

2 

3 

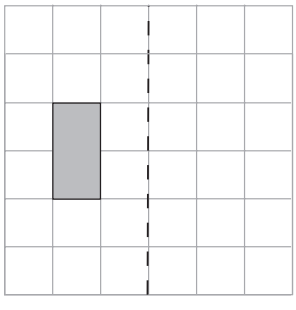
4 

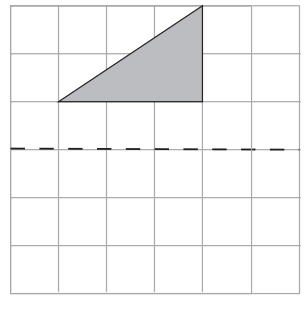
5 

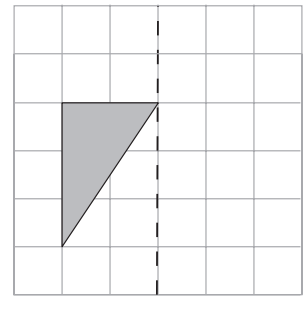
6 

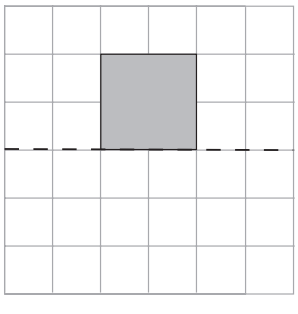
B

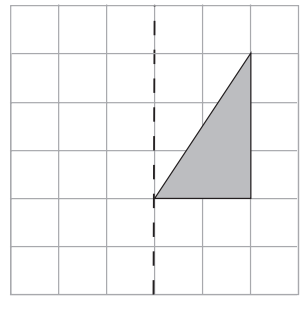
Draw the reflection of each figure in the dotted line. Use the grid squares to help you.

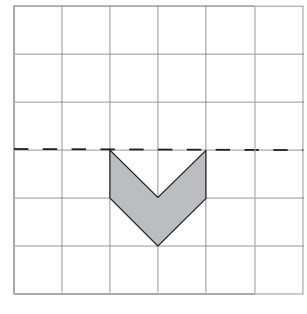
1 

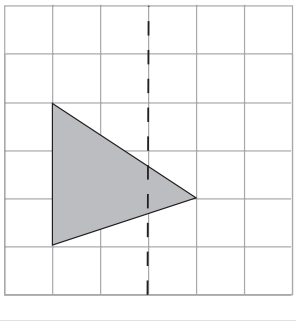
2 

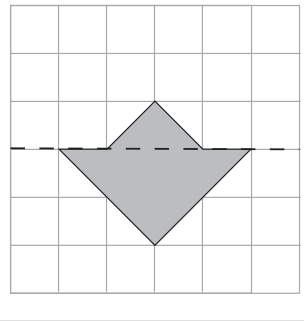
3 

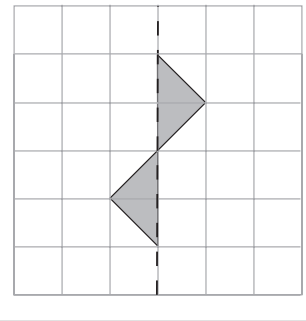
4 

5 

6 

7 

8 

9 

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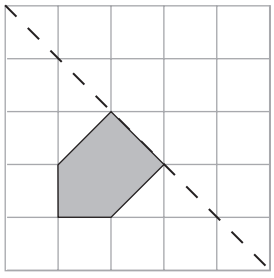
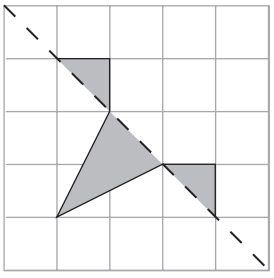
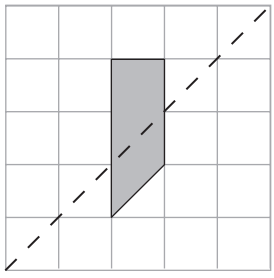
10



11

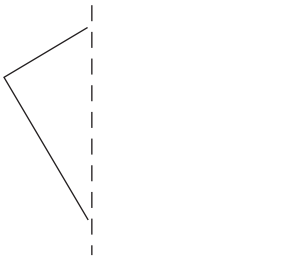


12



Complete these reflections. The dotted line should be an *axis of symmetry* of your completed diagram.

13



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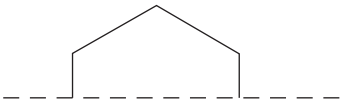
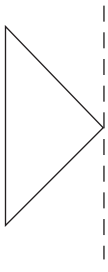


15

16



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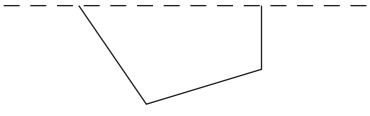

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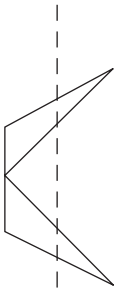
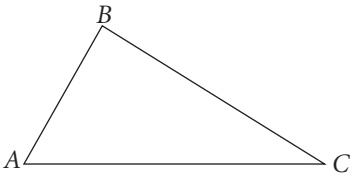
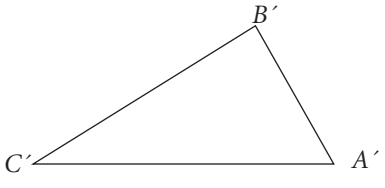


C

$A'B'C'$ is a reflection of the triangle ABC.





1 Measure the sides of each triangle to the nearest millimetre. Are matching sides the same length?

2 Using a ruler, mark the point X halfway between B and B'. Also mark the point Y, halfway between C and C'.

3 Draw the line of reflection passing through the points X and Y.

4 Are the points A,B,C and A',B',C' in the same rotational order (clockwise or anticlockwise)?

5 Complete this summary using the wordbank to fill in the missing words.

Under a reflection,

- a point and its image are the same _____ from the line of reflection.
- a line and its image are the _____ length.
- an _____ and its image are the same size.
- a figure and its _____ have the same area.
- a figure and its image are _____.
- the _____ order of the vertices of a figure is reversed.

Wordbank: *angle, congruent, distance, image, rotational, same.*
(Each word may be used once only.)

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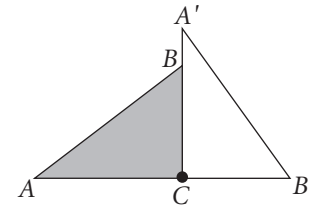
Rotation 14.1

The movement of a plane shape to a new position by spinning it around a fixed point is called a *rotation.*
When we describe a rotation, we have to state the size of the rotation (the number of degrees), the direction (clockwise or anticlockwise) and the *centre of rotation* (the fixed point which does not move).
Under rotation, the figure remains the same shape and size as the original figure.

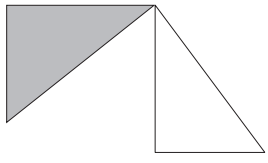
Example:
The triangle *ABC* has been rotated 90° in a clockwise direction about the point *C.*
A′ is the image of the point A.



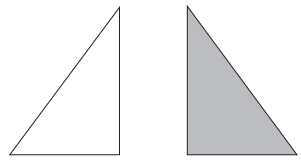
A

Are the following transformations rotations?

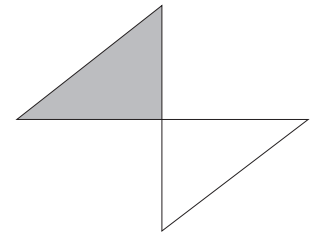
1



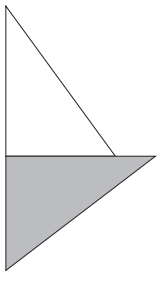
2



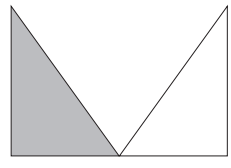
3



4



5



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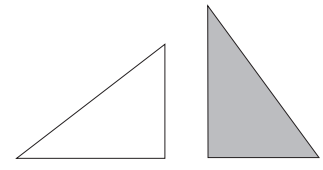
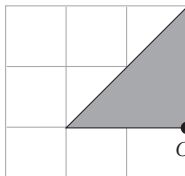


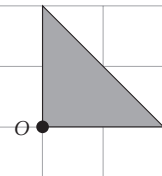
B

Rotate each figure through the angle indicated. The point *O* is the centre of rotation in each diagram.

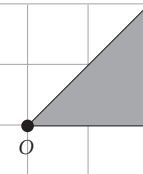
1 90° clockwise



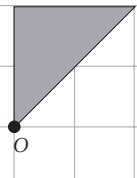
2 180° clockwise



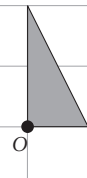
3 90° anticlockwise



4 180° anticlockwise



5 90° clockwise



6 180° clockwise

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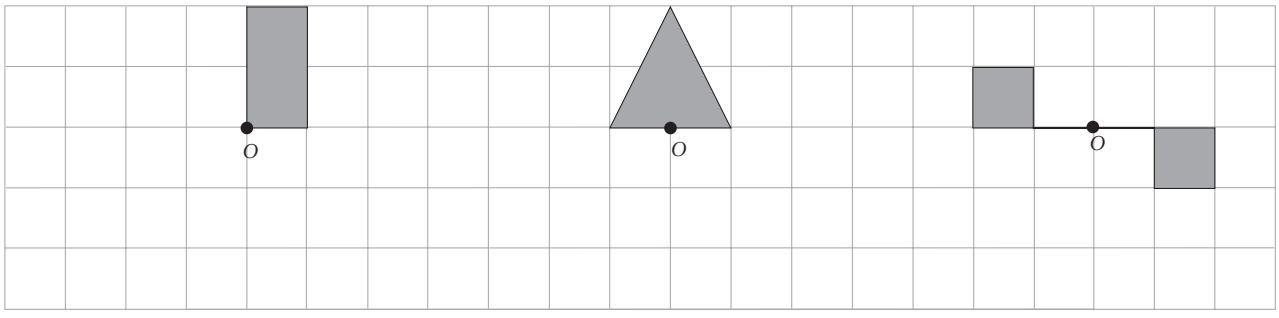
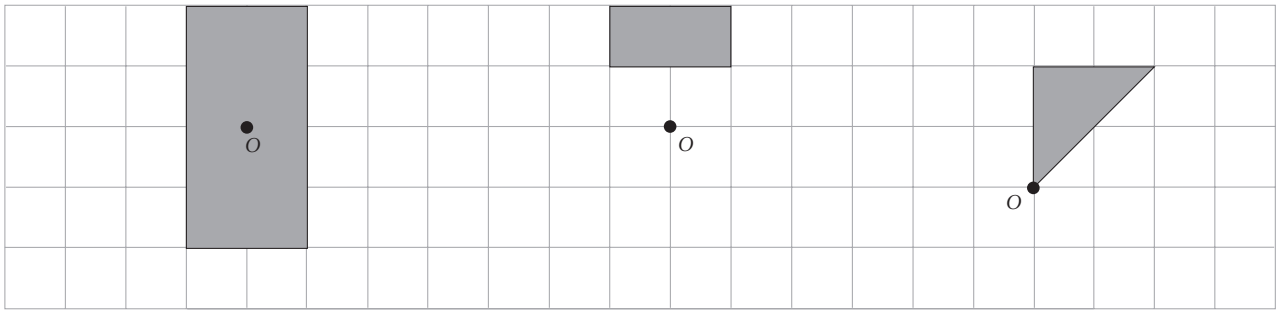
7 90° anticlockwise



8 180° anticlockwise

9 90° clockwise

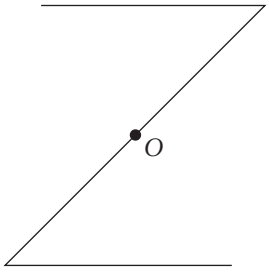
10 90° anticlockwise



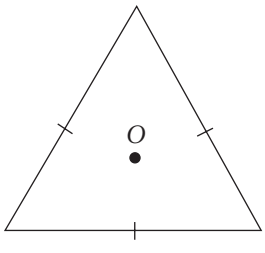
11 270° clockwise

12 45° clockwise

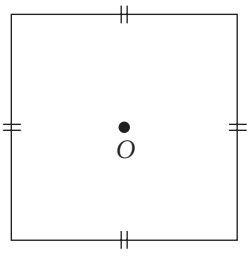
13 90° clockwise



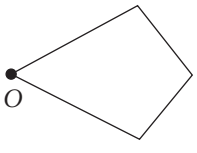
14 180° anticlockwise



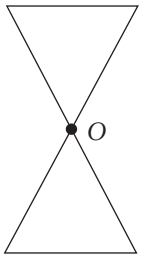
15 45° clockwise



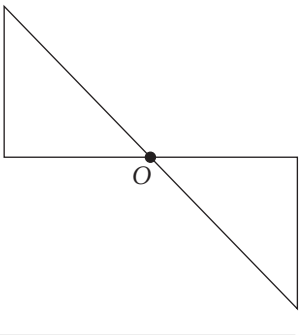
16 180° clockwise



17 90° anticlockwise



18 90° clockwise



C Complete this summary, using the wordbank to fill in the missing words.

Under a rotation,

- a point and its _____ are the same distance from the centre of rotation.
- a line and its image are the same _____.
- an angle and its image are the _____ size.
- a figure and its image have the same _____.
- a _____ and its image are congruent.
- the rotational order of the vertices of a figure is _____.

Wordbank : *area, figure, image, length, same, unchanged.*
(Each word may be used once only.)

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Congruent figures 15.1

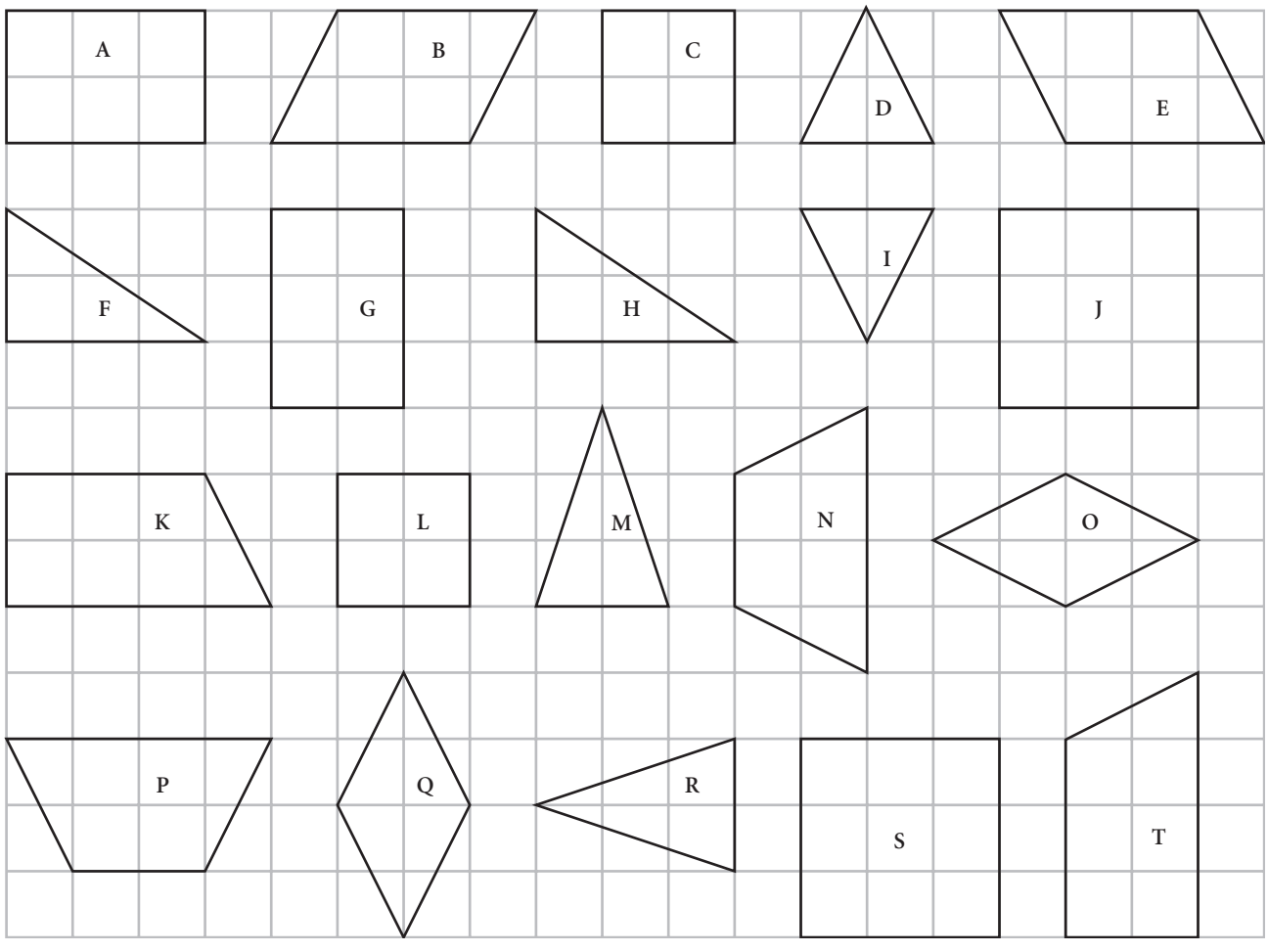
A

Congruent figures are identical in shape and size.
Two figures are congruent if:

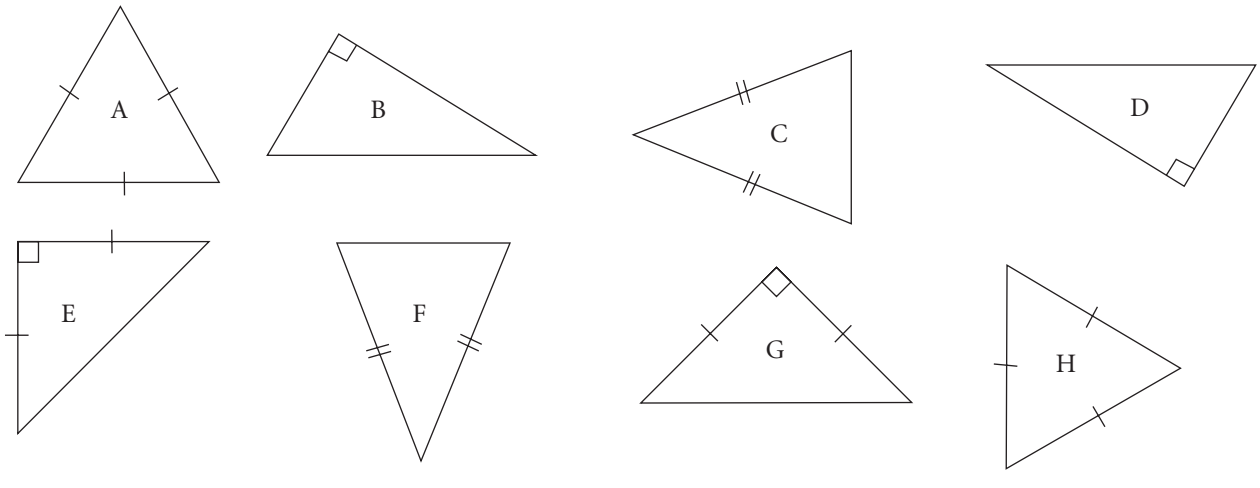
- one figure can be moved (translated or rotated) so that it will superimpose on the other; or
- one figure is the mirror image of the other; or
- congruency can be proved by deductive reasoning.

The symbol ≡ means 'is congruent to'.

1 Match the following figures into pairs of congruent shapes.



2 Match the following triangles into congruent pairs.



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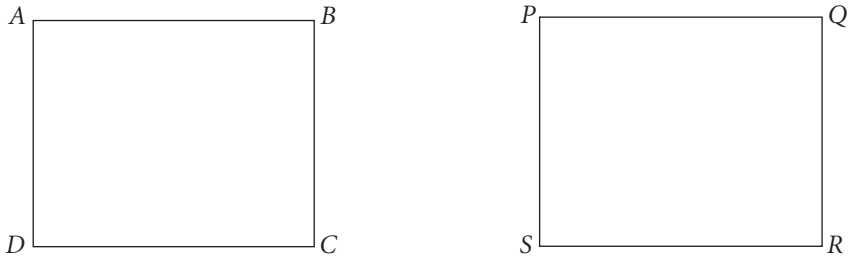
B

Use a ruler to measure the length of each side of these figures to the nearest millimetre. Then use a protractor to measure each angle.

1 Two rectangles:

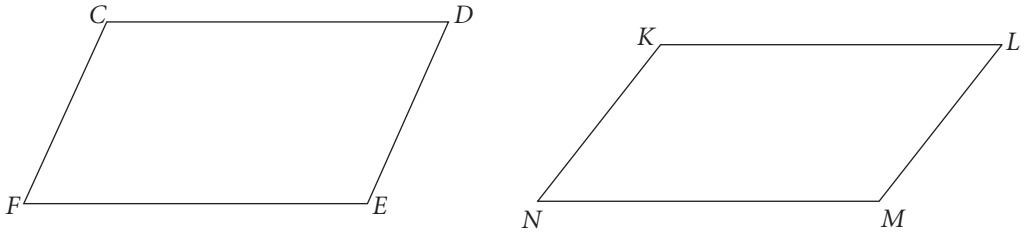


a Do these rectangles have the same length? (Does $AB = PQ$?)

b Do they have the same breadth? (Does $AD = PS$?)

c Are these two rectangles congruent?

2 Two parallelograms:

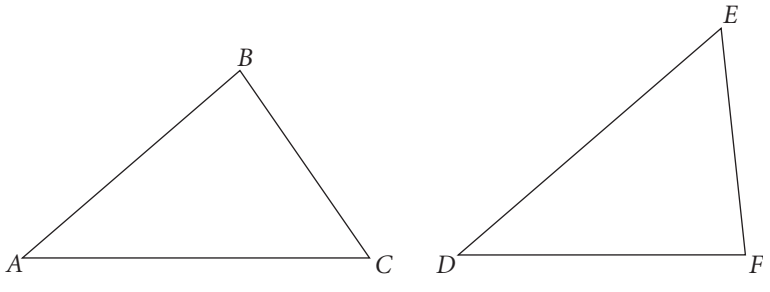


a Do these parallelograms have the same length? (Does $FE = NM$?)

b Are the sides FC, DE, NK and LM the same length?

c Are these two parallelograms congruent?

3 Two triangles:



a Are there three pairs of equal sides?

b Are matching angles the same size?

c Are these two triangles congruent?

C

1 If each member of the class constructed a square with sides 6 cm long, should all the squares be congruent? Why?

2 If each member of the class constructed a rectangle with length 5 cm and breadth 4 cm, should all the rectangles be congruent? Why?

3 If each member of the class constructed a triangle with sides measuring 6 cm, 8 cm and 9 cm, should all the triangles be congruent? Why?

4 If each member of the class constructed a parallelogram with one pair of parallel sides measuring 6 cm and the other pair of parallel sides measuring 4 cm, should all the parallelograms be congruent? Why?

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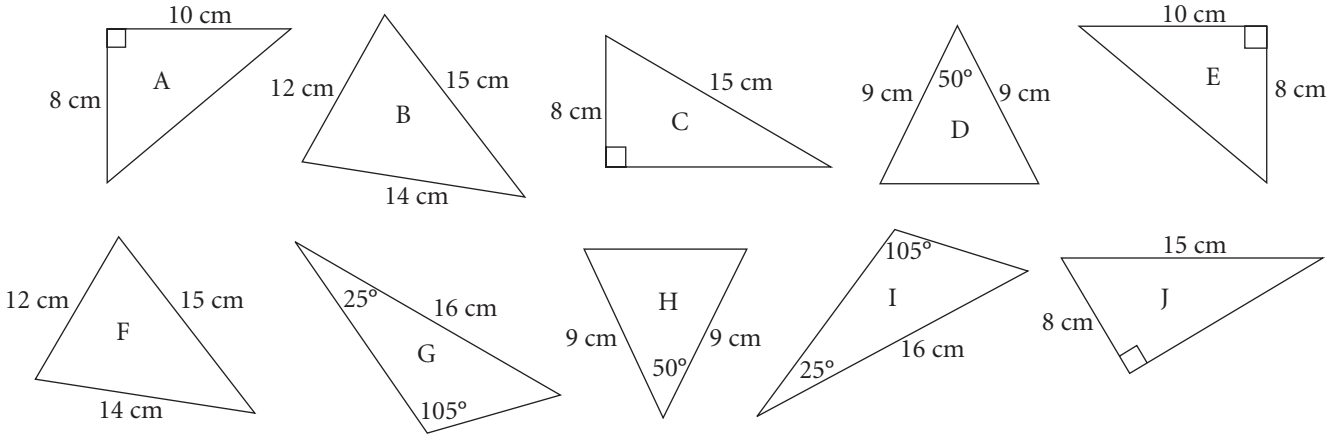
Congruent triangles 16.1

A

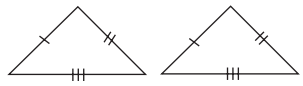
Match the following triangles into congruent pairs.



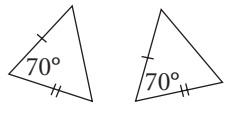
B

There are four tests to show that two triangles are congruent.

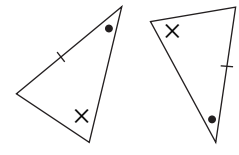
- ***SSS test:*** The three sides of one triangle are equal to the three sides of the other triangle.



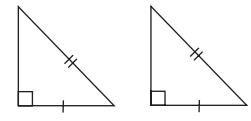
- ***SAS test:*** Two sides and the included angle of one triangle are equal to the matching two sides and the included angle of the other triangle.



- ***AAS test:*** Two angles and one side of one triangle are equal to two angles and the matching side of the other triangle.

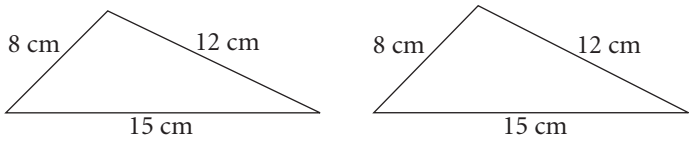


- ***RHS test:*** In two right-angled triangles, the hypotenuses are equal and one other pair of sides are also equal.



For each pair of congruent triangles, state which test (SSS, SAS, AAS or RHS) can be applied using the given information.

1



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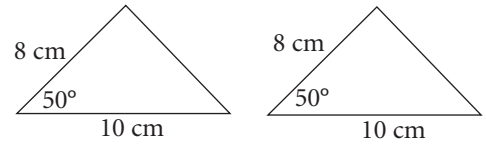

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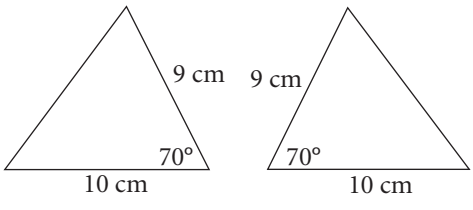

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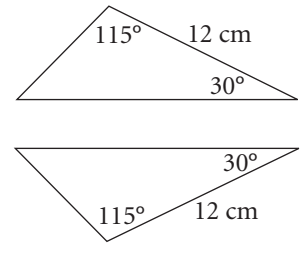

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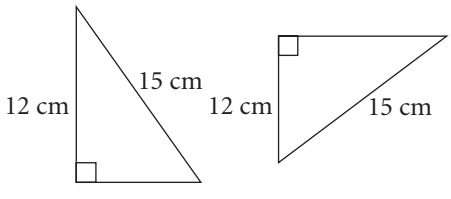

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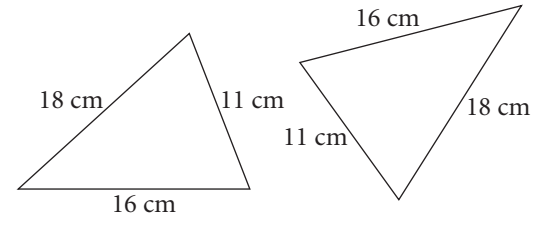

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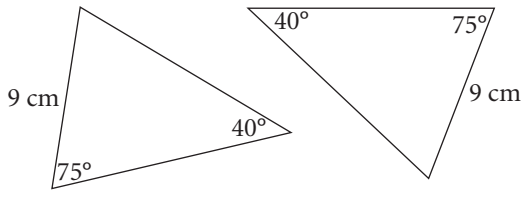
7



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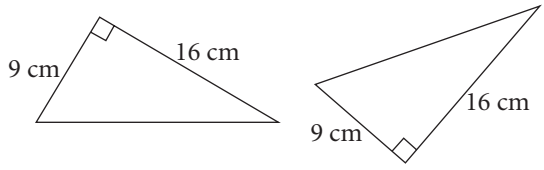
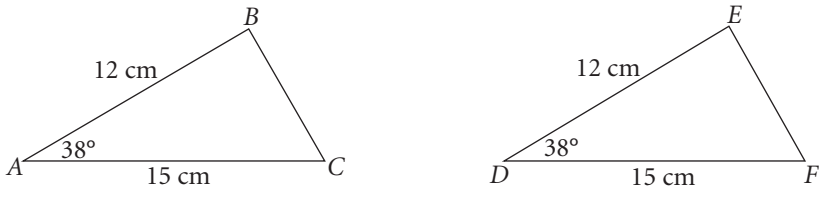


The triangles *ABC* and *DEF* are congruent.



9 Which test can be applied to show that the triangles are congruent?

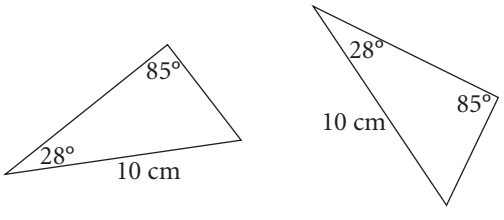
10 Complete these statements to indicate pairs of equal sides and pairs of equal angles for the triangles *ABC* and *DEF* above.

a $AB =$ _____ **b** $AC =$ _____ **c** $BC =$ _____

d $\angle BAC =$ _____ **e** $\angle ABC =$ _____ **f** $\angle BCA =$ _____

C Are these pairs of triangles congruent? Sometimes you will need to calculate the size of the third angle before making your decision. For each pair of congruent triangles, state which test you were able to apply.

1



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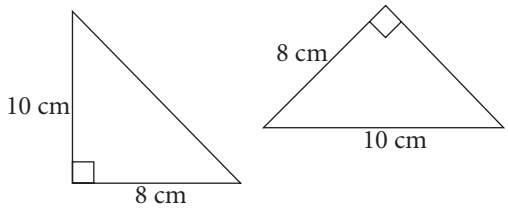

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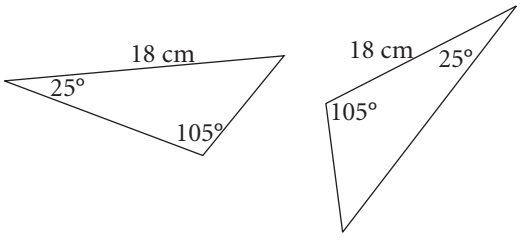

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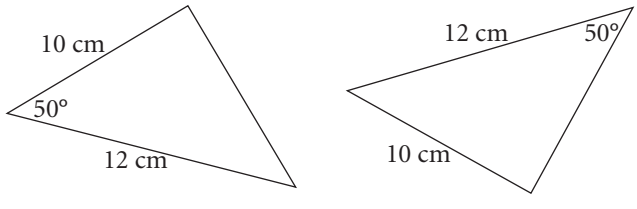

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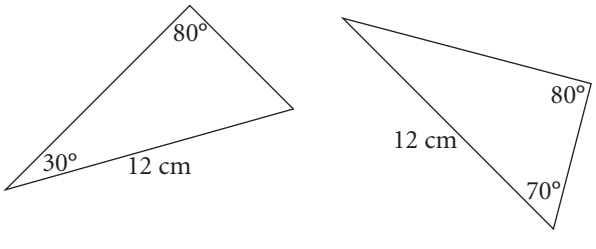

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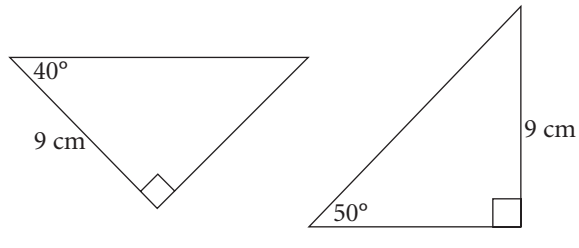

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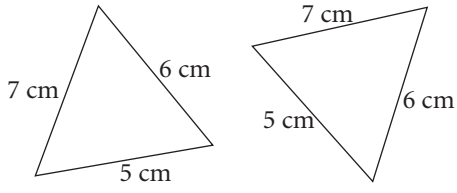
Writing congruence proofs 17.1

A For each pair of congruent triangles, state which test (SSS, SAS, AAS or RHS) can be applied using the given information.

1



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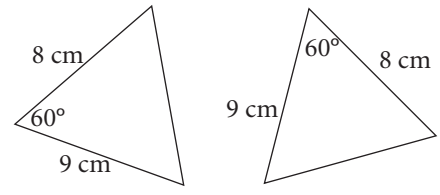

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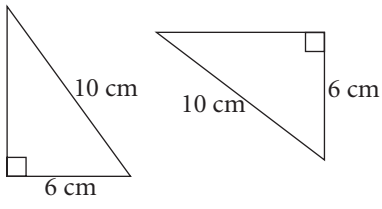

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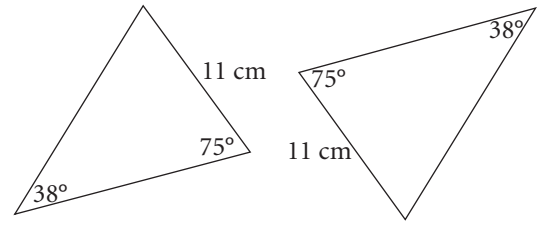
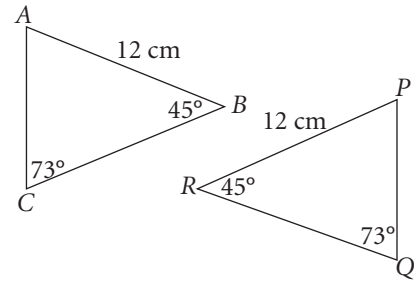


5 Which test can be used to explain why the triangles *ABC* and *PQR* are congruent?



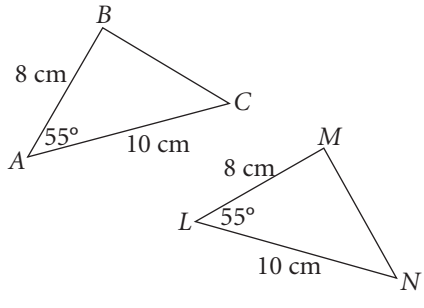
6 Match the pairs of equal sides and equal angles in triangles *ABC* and *PQR*, and complete these statements:

a $AB =$ _____ **b** $AC =$ _____ **c** $BC =$ _____

d $\angle ABC =$ _____ **e** $\angle ACB =$ _____ **f** $\angle BAC =$ _____

B

A proof is a formal explanation, using data that is known to be true. This is an example of a proof that the two triangles *ABC* and *LMN* are congruent.

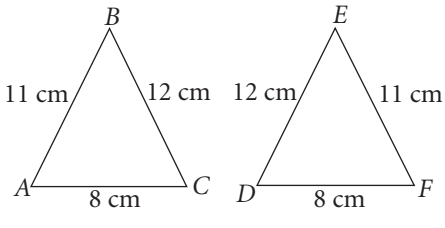


Example: In the triangles *ABC* and *LMN*,
S : $AB = LM$ (both 8 cm)
A : $\angle BAC = \angle MLN$ (both 55°)
S : $AC = LN$ (both 10 cm)
$\therefore \Delta ABC \equiv \Delta LMN$ (SAS test)

Note: Using the SAS test, we state the data in the order *side, angle, side*. A brief reason is given in parentheses for each statement.

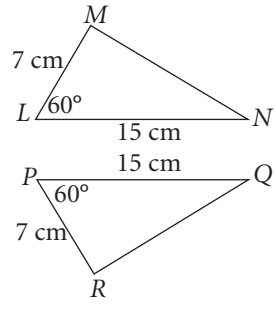
Complete these congruence proofs.

1



In the triangles *ABC* and *DEF*,
S : $AB =$ _____ (both 11 cm)
S : $BC =$ _____ (both _____)
S : $AC =$ _____ (_____)
$\therefore \Delta ABC \equiv \Delta DEF$ (_____ test)

2



In the triangles *LMN* and *PQR*,
S : $LM =$ _____ (both _____)
A: $\angle MLN =$ _____ (both _____)
S : $LN =$ _____ (_____)
$\therefore \Delta LMN \equiv \Delta$ _____ (_____ test)

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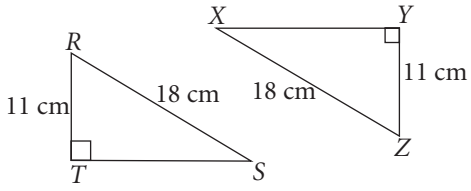
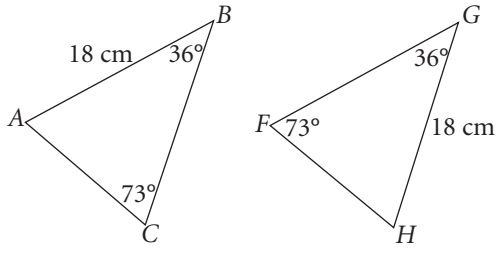
3



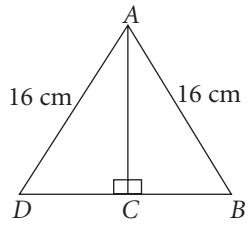
In the triangles *RST* and _____,
R : $\angle RTS =$ _____ (both 90°)
H : $RS =$ _____ (_____)
S : _____ = _____ (both 11 cm)
$\therefore \Delta RST \equiv$ _____ (_____ test)

4



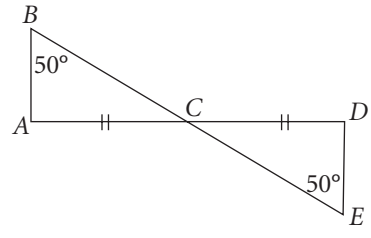
In the triangles _____ and _____,
A : $\angle ABC =$ _____ (_____)
A : _____ = _____ (_____)
S : $AB =$ _____ (both 18 cm)
$\therefore \Delta ABC \equiv$ _____ (_____)

5



In the triangles *ADC* and *ABC*,
R : $\angle ACD =$ _____ (_____)
H : _____ = _____ (_____)
S : *AC* is common
$\therefore \Delta$ _____ $\equiv \Delta$ _____ (_____)

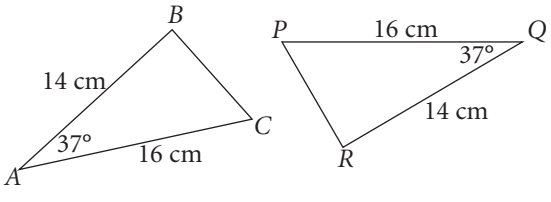
6



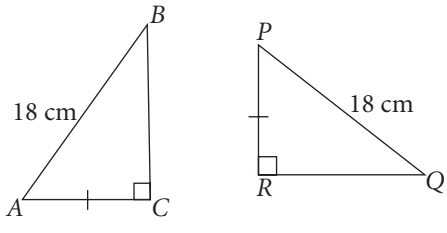
In the triangles *ABC* and *CDE*,
A : $\angle ABC =$ _____ (_____)
A : $\angle ACB =$ _____ (vertically opposite angles)
S : $AC =$ _____ (given)
$\therefore$ _____ $\equiv$ _____ (_____ test)

C Use the information on each diagram to prove that $\Delta ABC \equiv \Delta PQR$.

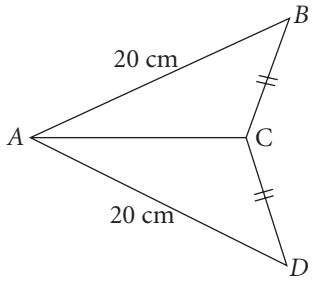
1



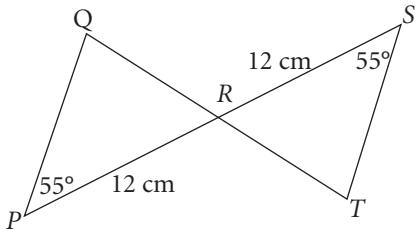
2



3 Prove that $\Delta ABC \equiv \Delta ADC$.



4 **a** Why does $\angle PRQ = \angle SRT$?
b Prove that $\Delta PRQ \equiv \Delta SRT$



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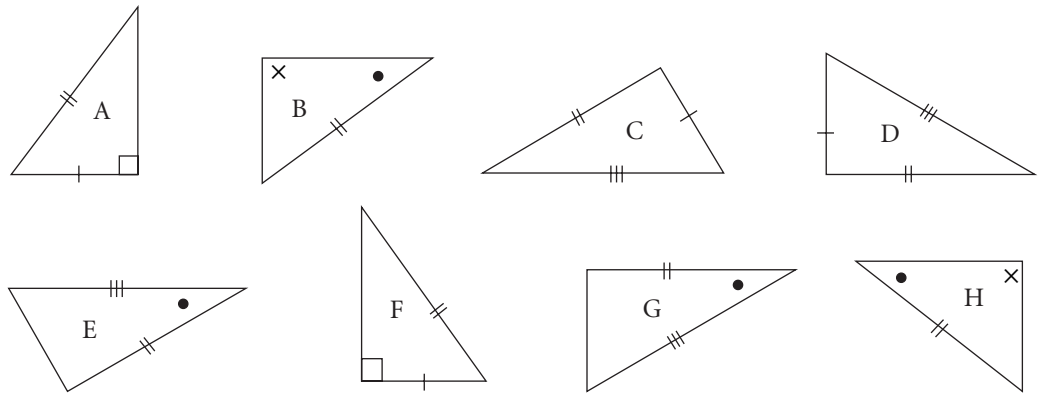
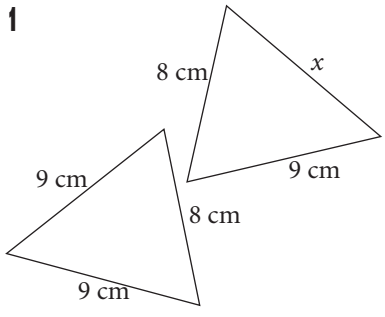
Applying congruence 18.1

A Match the following triangles into congruent pairs, and state which congruency test (*SSS*, *SAS*, *AAS*, or *RHS*) you applied. Equal sides and equal angles are indicated using the same symbol.



B Given each pair of triangles is congruent, find the value of each pronumeral.

1



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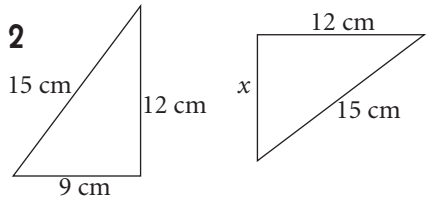

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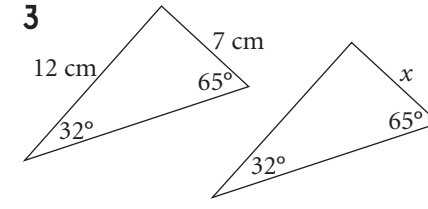

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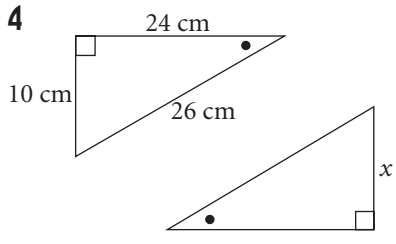
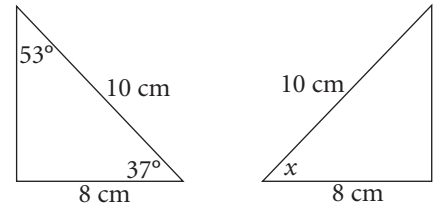


5



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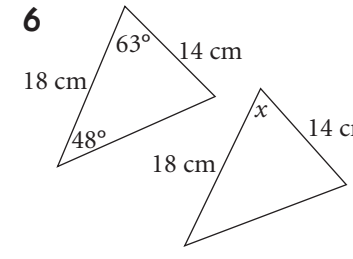

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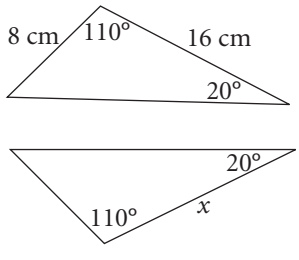

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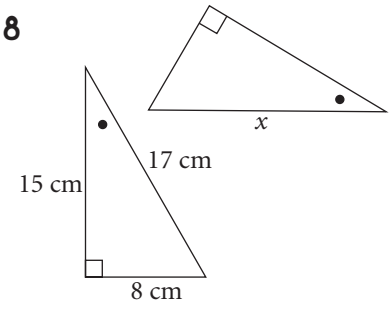

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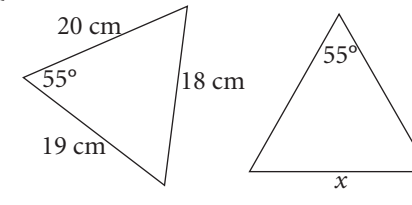

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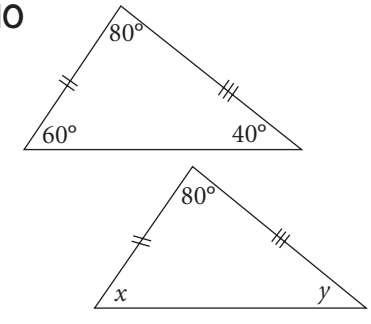

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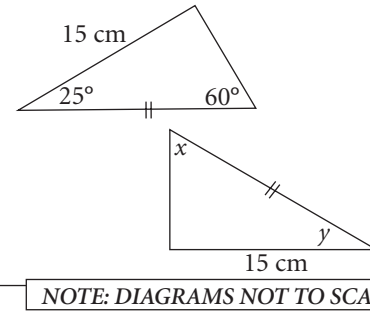

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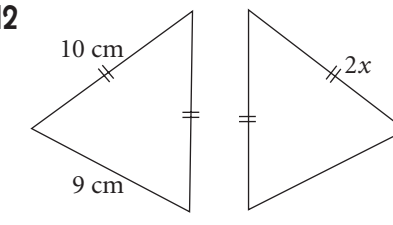

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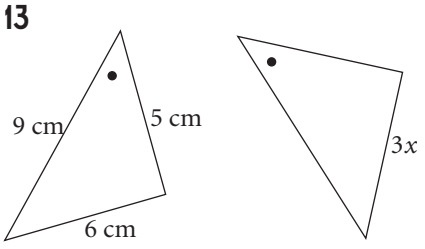
13



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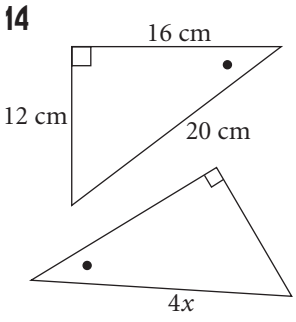
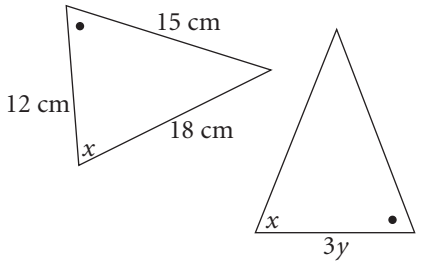
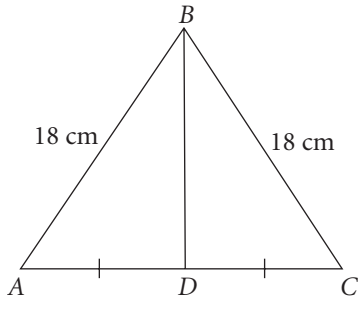


15

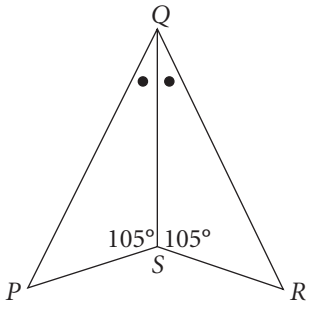


C

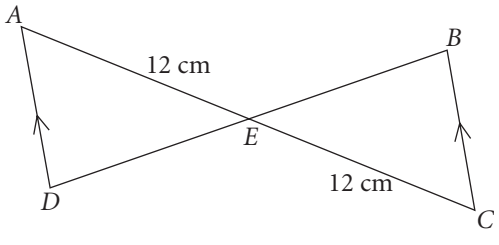
1 Prove that $\Delta ABD \equiv \Delta CBD$.



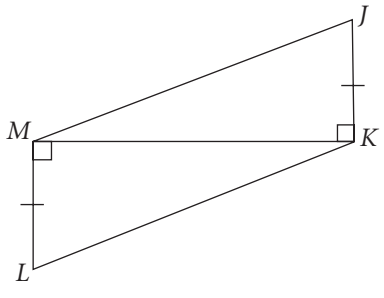
2 Prove that $\Delta PQS \equiv \Delta RQS$.



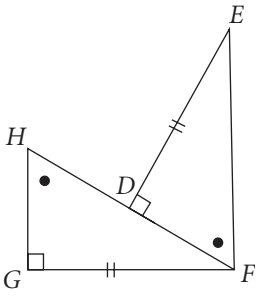
3 Prove that $\Delta ADE \equiv \Delta CBE$.



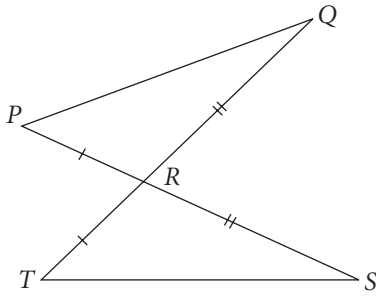
4 Prove that $\Delta JKM \equiv \Delta LMK$.



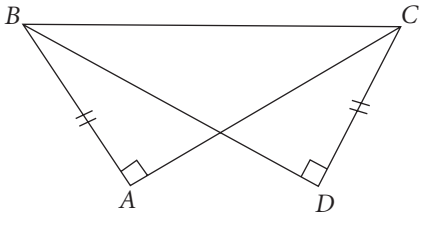
5 Prove that $\Delta DEF \equiv \Delta FGH$.



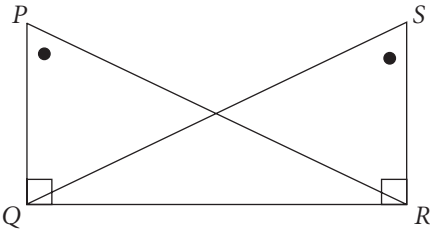
6 Prove that $\Delta PQR \equiv \Delta STR$.



7 Prove that $\Delta ABC \equiv \Delta DBC$.



8 PQ and SR are drawn at right angles to the line QR. Prove that $\Delta PQR \equiv \Delta QRS$.



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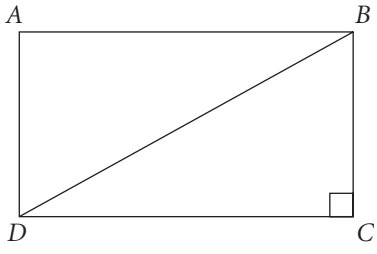
Geometric proofs 19.1

Triangles are perhaps the most common shape found in theoretical and practical problems in geometry. The proof of many of the properties of geometric shapes requires you to prove that two triangles are congruent as one step in your reasoning.

A

Complete these proofs by filling in the missing statements.

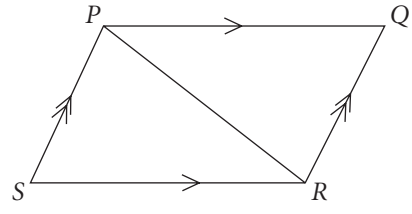
1 *ABCD* is a rectangle.
The diagonal *BD* has been drawn.



Prove that $\Delta ABD \equiv \Delta CBD$.

In the triangles *ABD* and *CBD*,
$AB =$ _____ (opposite sides of a rectangle are equal)
$\angle DAB =$ ____ (_____)
$AD =$ _____ (_____)
$\therefore \Delta ABD \equiv \Delta CBD$ (_____ test)

2 *PQRS* is a parallelogram.
The diagonal *PR* has been drawn.



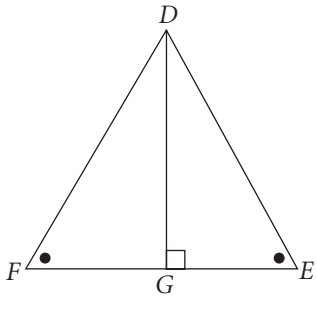
Prove that:

a $\Delta PQR \equiv \Delta PSR$

b $PQ = SR$ and $PS = QR$

In the triangles *PQR* and *PSR*,
$\angle QPR =$ _____ (equal alternate angles)
$\angle PRQ =$ _____ (_____)
PR is _____
$\therefore \Delta PQR \equiv \Delta PSR$ (_____ test)
$\therefore PQ =$ _____ and $PS =$ _____
(matching sides of the congruent triangles)

3 In the triangle *DEF*, $\angle E = \angle F$.
DG has been drawn perpendicular to the side *EF*.



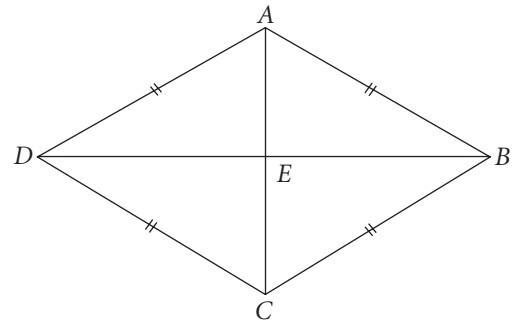
Prove that:

a $\Delta DFG \equiv \Delta DEG$

b $DF = DE$

In the triangles *DFG* and *DEG*,
$\angle DFG =$ _____ (given)
$\angle DGF =$ _____ (_____)
DG is _____
$\therefore \Delta DFG \equiv \Delta DEG$ (_____ test)
$\therefore DF = DE$ (_____)

4 *ABCD* is a rhombus. The diagonals *AC* and *BD* meet at the point *E*.



Prove that:

a $\Delta AED \equiv \Delta AEB$

b $\angle AEB = 90°$

In the triangles *AED* and *AEB*,
$AD =$ _____ (_____)
$\angle DAE = \angle BAE$ (_____)
AE is _____
$\therefore \Delta AED \equiv \Delta AEB$ (_____ test)
$\angle AED = \angle$_____ (_____)

However,
$\angle AED + \angle AEB =$ _____ (*BD* is a straight line)
$\therefore \angle AEB = 90°$

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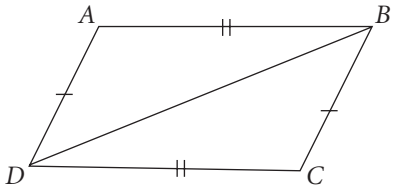
B

1 *ABCD* is a quadrilateral in which the opposite sides are the same length.
Prove that:

a $\Delta ABD \equiv \Delta CBD$

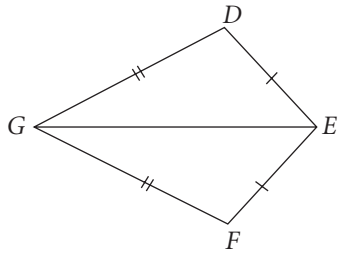
b $\angle BAD = \angle BCD$



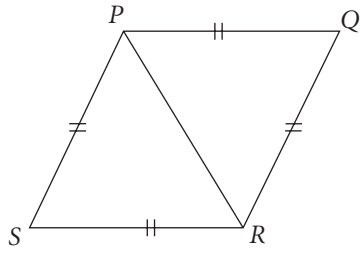
2 *DEFG* is a kite where $DE = EF$ and $GD = GF$.
Prove that:

a $\Delta DEG \equiv \Delta FEG$

b $\angle DGE = \angle FGE$



3 *PQRS* is a rhombus.

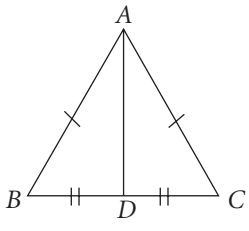


a Prove that $\Delta PQR \equiv \Delta PSR$.

b Hence prove that $\angle QPR = \angle SPR$ and $\angle QRP = \angle SRP$.

c Complete the statement:
'The diagonals of a rhombus _____ the angles of the rhombus.'

4 *ABC* is an isosceles triangle ($AB = AC$)
D is the midpoint of *BC* ($BD = DC$).



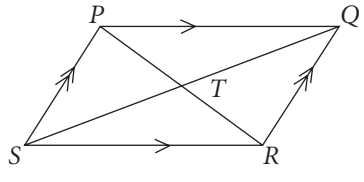
a Prove that $\Delta ABD \equiv \Delta ACD$.

b Explain why $\angle ABD = \angle ACD$.

c Complete the statement:
'In an isosceles triangle, the angles opposite the equal sides are _____.'

C

1 *PQRS* is a parallelogram.
The diagonals *PR* and *QS* meet at the point *T*.

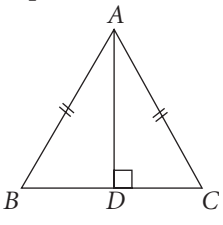


a Prove that $\Delta PQT \equiv \Delta RST$.

b Prove that $PT = TR$ and $ST = TQ$.

c What property of a parallelogram have you proved?

2 *ABC* is an isosceles triangle.
AD has been drawn perpendicular to *BC*.

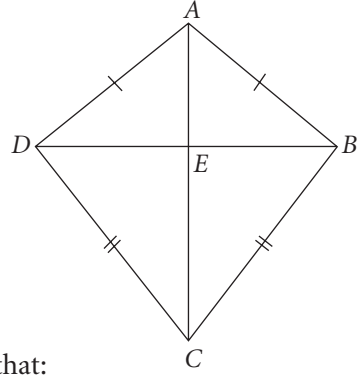


Prove that

a $\Delta ABD \equiv \Delta ACD$

b *D* is the midpoint of *BC* (i.e. $BD = DC$).

3 *ABCD* is a kite.
($AB = AD$, $BC = DC$)
The diagonals *AC* and *BD* meet at the point *E*.



Prove that:

a $\Delta ADC \equiv \Delta ABC$

b $\angle DAE = \angle BAE$

c $\Delta ADE \equiv \Delta ABE$

d the diagonals *AC* and *BD* meet at right angles (i.e. $AED = 90°$).

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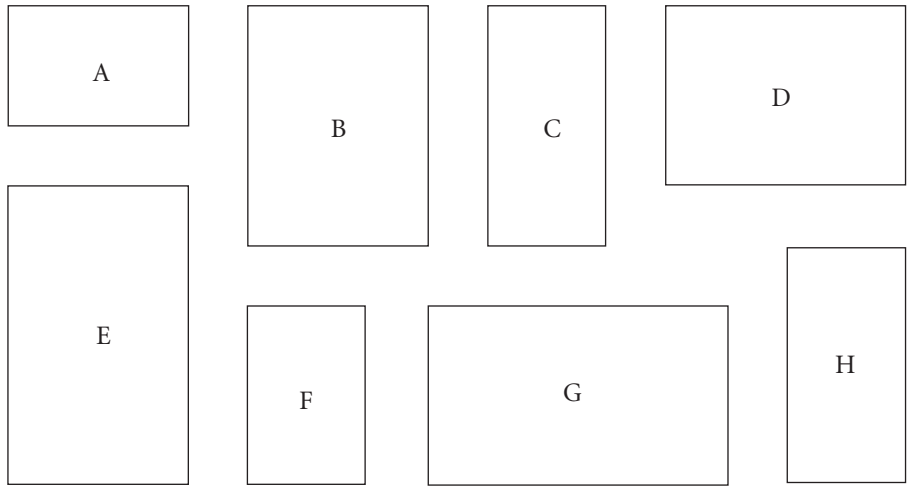
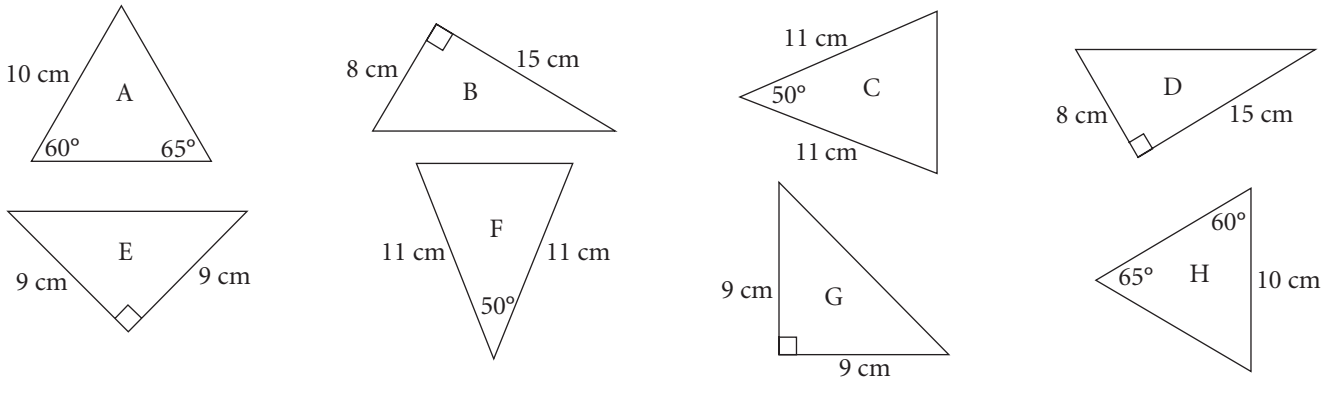
Review — congruent figures 20.1

A

1 Match the following rectangles into congruent pairs.



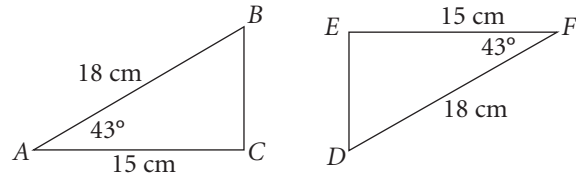
2 Match the following triangles into congruent pairs.



3 The two triangles *ABC* and *DEF* are congruent.
Match pairs of equal sides and equal angles, and complete these statements:

a $AB =$ _____ **b** $BC =$ _____ **c** $AC =$ _____

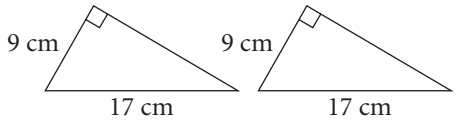
d $\angle BAC =$ _____ **e** $\angle ABC =$ _____ **f** $\angle BCA =$ _____



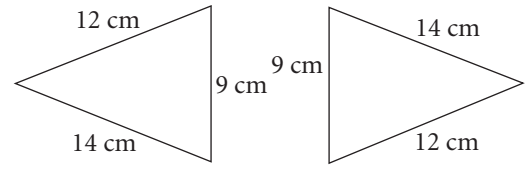
B

For each pair of congruent triangles, state which test (SSS, SAS, AAS, or RHS) can be applied using the given information.

1



2



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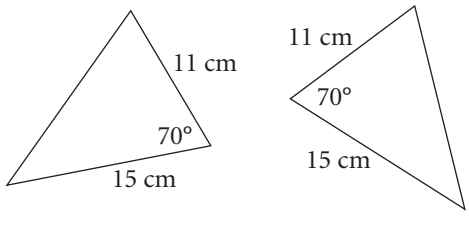

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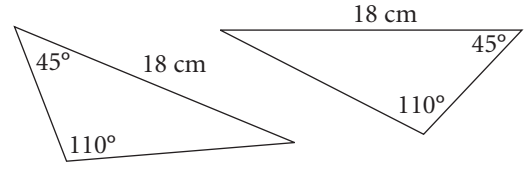

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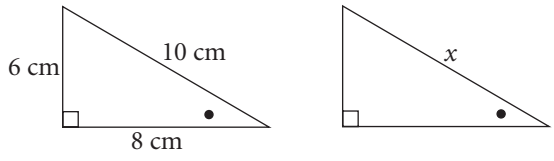
Find the value of the pronumeral x in each pair of congruent triangles.

5



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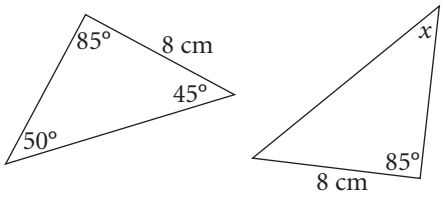

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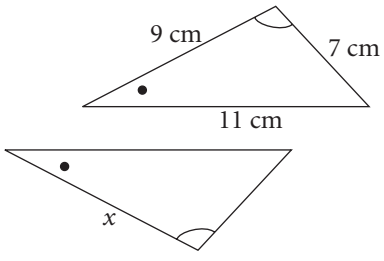

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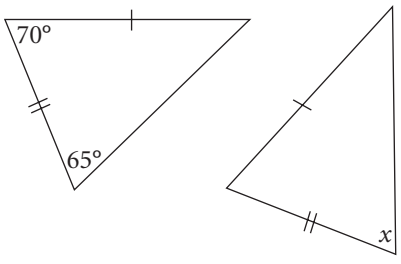
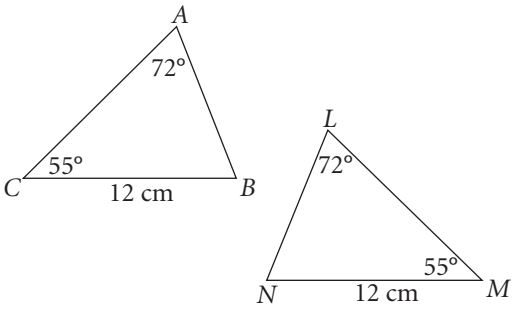
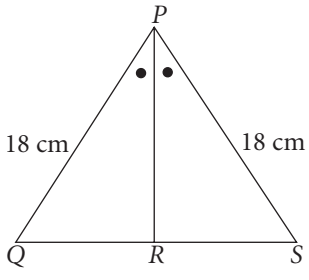


9 Complete this proof that $\Delta ABC \equiv \Delta LMN$.



In the triangles ABC and LMN,
A: $\angle ACB =$ _____ (_____)
A: $\angle BAC =$ _____ (_____)
S: $BC =$ _____ (_____)
$\therefore \Delta ABC \equiv \Delta LMN$ (_____)

10 Complete this proof that $\Delta PQR \equiv \Delta PSR$.

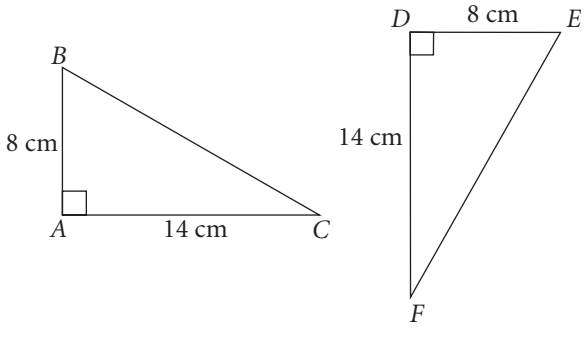


In the triangles PQR and PSR,
$PQ =$ _____ (_____)
$\angle QPR =$ _____ (_____)
PR is _____
$\therefore \Delta PQR \equiv \Delta PSR$ (_____ test)

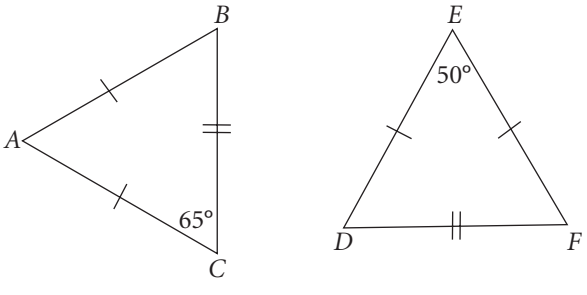
C

Use the information given on each pair of diagrams to prove that $\Delta ABC \equiv \Delta DEF$.

1



2



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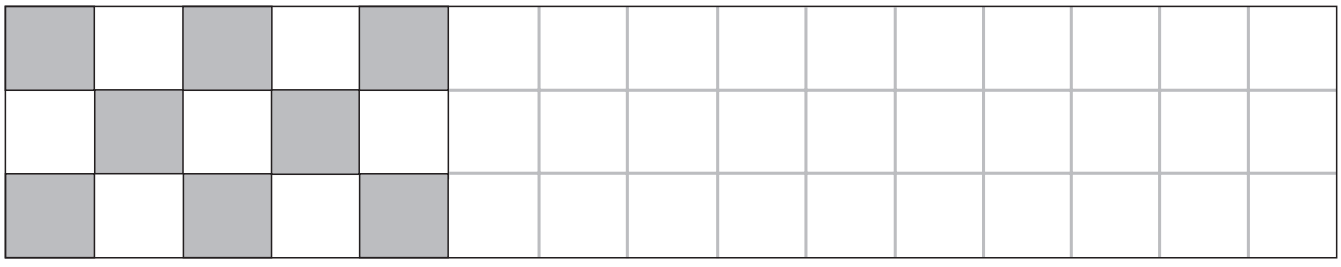
Tessellations 21.1

A

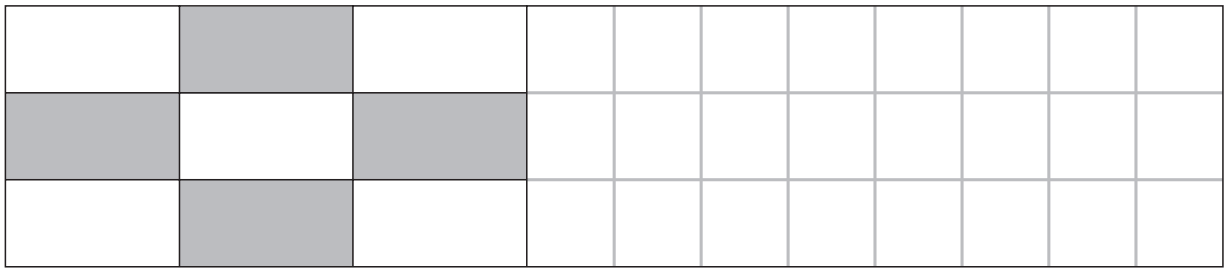
A tessellation is where we completely cover a surface by repeating a pattern of shapes, leaving no gaps. Tessellations can be formed using a translation, rotation or reflection of a simple plane shape.

Complete these tessellations to cover the area shown.

1



2



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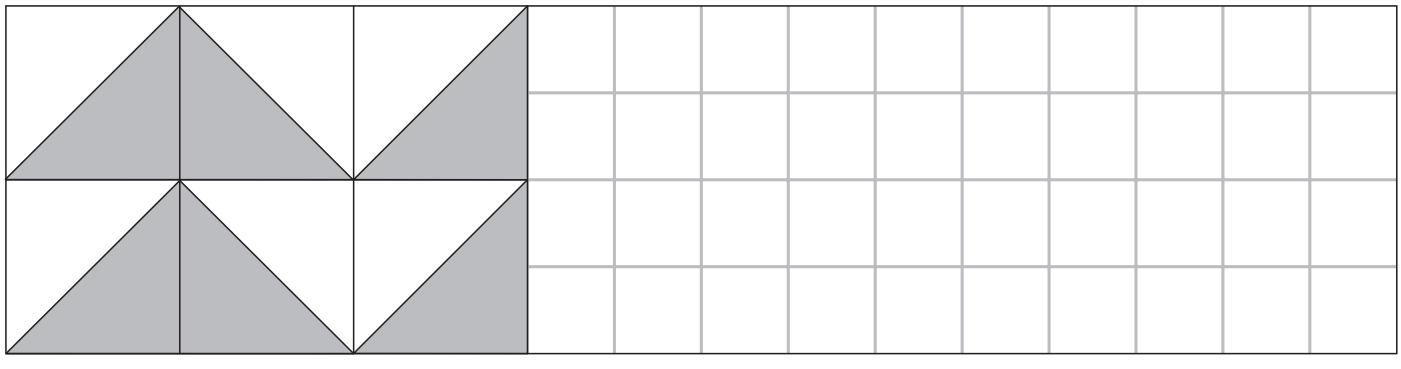
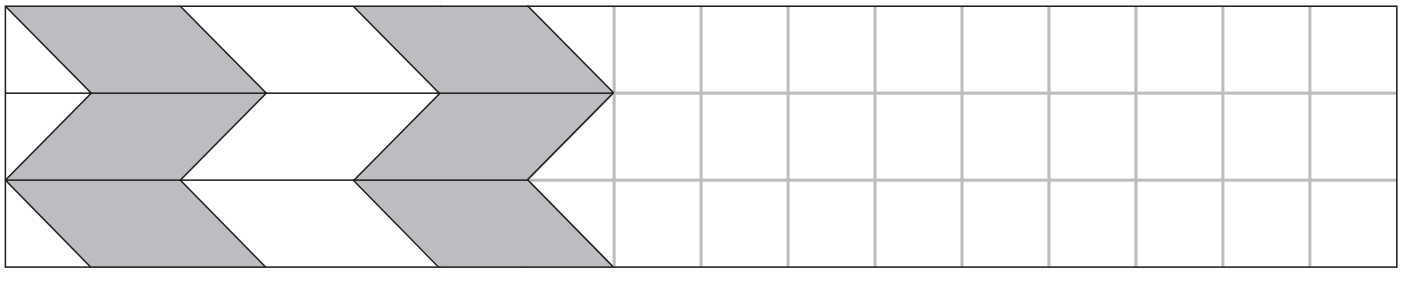
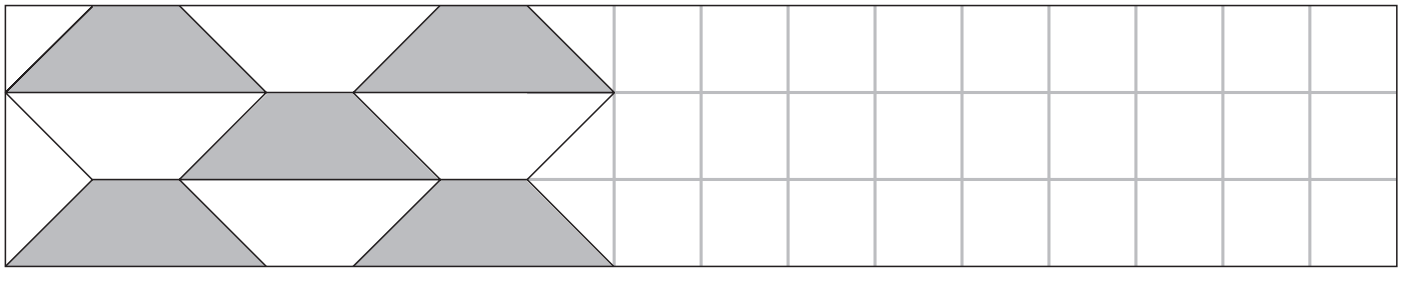


4



5



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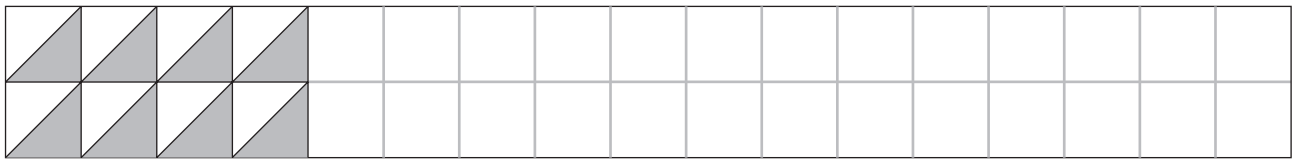
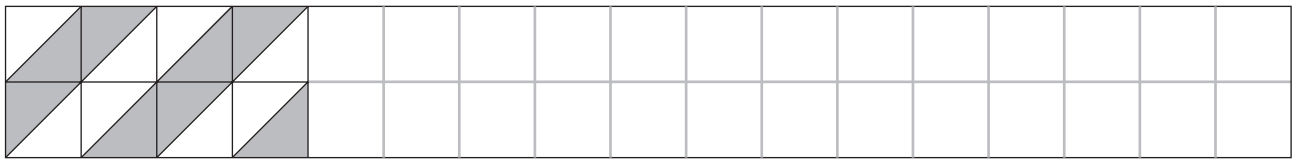
B Different patterns can be achieved using the same basic shape (or tile) but different transformations. These tessellations all use the same basic square tile: ◪. Complete each tessellation across the page using the nominated transformation.

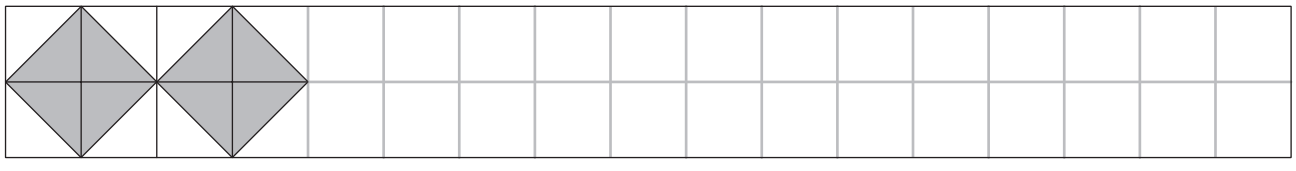
1 translation



2 180° rotation (the midpoint of the edge is the centre of rotation)

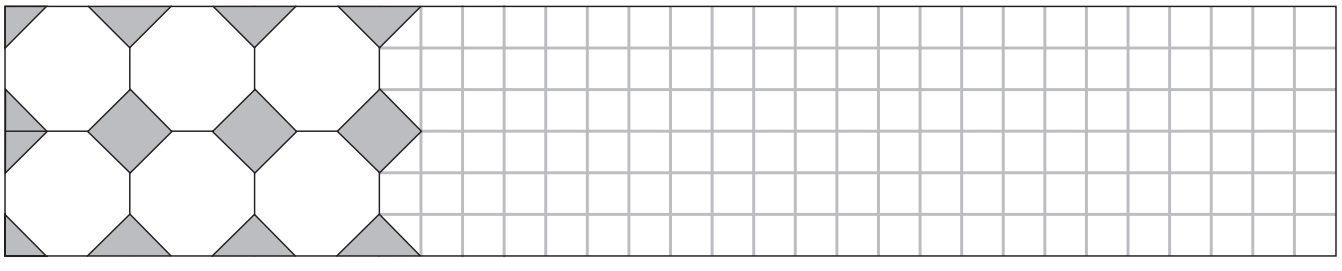


3 reflection

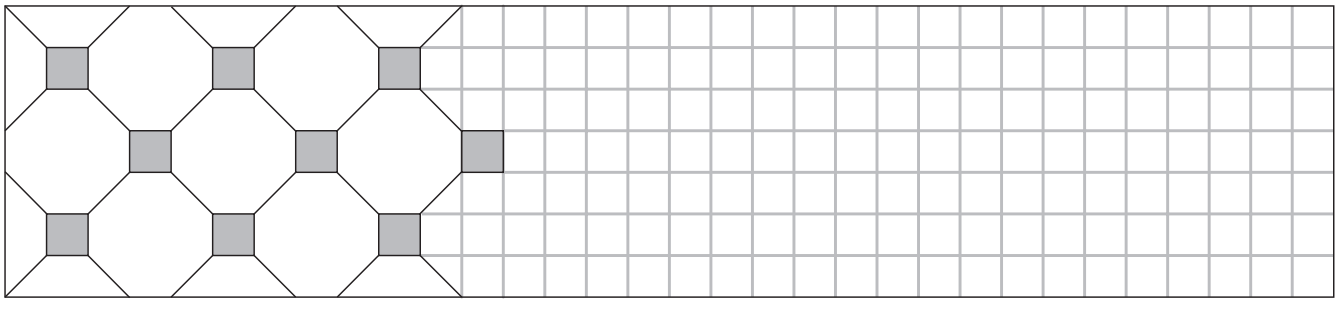


Many tessellations use a composite figure of two or more shapes as the basic repeating unit. Complete these tessellations to cover the area shown.

4



5



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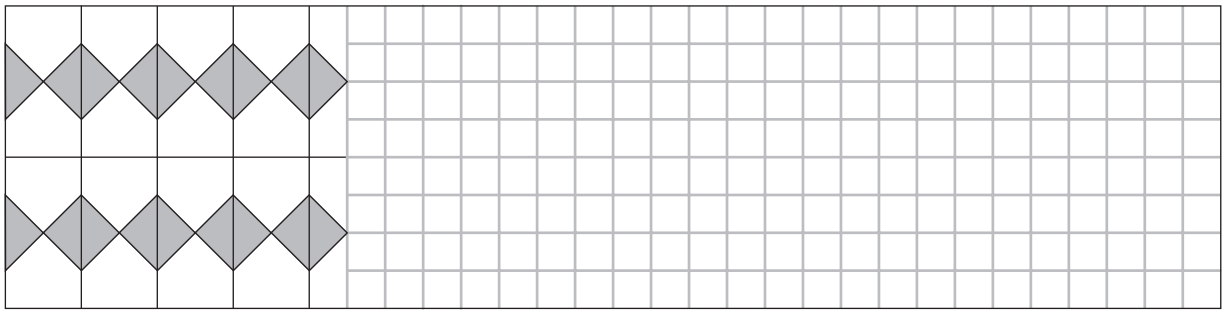
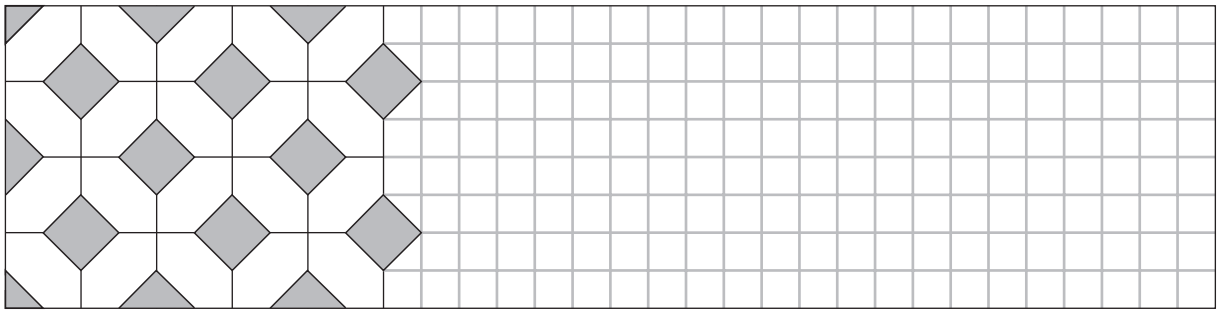
6

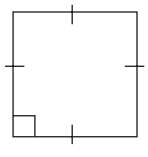


7



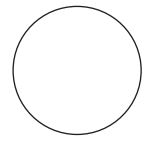
C Some plane shapes will tessellate (cover a surface without leaving gaps) while others will leave gaps. Will the following shapes tessellate? Draw examples on a piece of paper using a template, such as a Math-o-mat or Mathaid, to help make your decisions.

1 a square

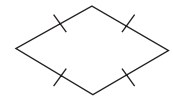


2 an equilateral triangle

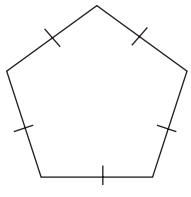
3 a circle



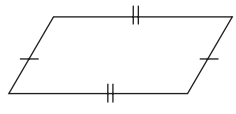
4 a rhombus



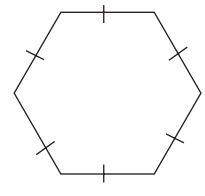
5 a regular pentagon



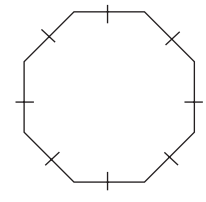
6 a parallelogram



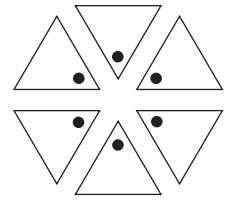
7 a regular hexagon



8 a regular octagon



We will now investigate why some shapes *will* tessellate.



9 What is the size of each angle in an equilateral triangle?

10 Could six equilateral triangles be fitted together at the one point?

11 What would the six angles at this point add up to?

12 What is the name of a regular quadrilateral?

13 Could four squares be positioned to fit together at one point?

14 What would the four angles at this point add up to?

15 Could three regular hexagons be positioned to fit together at one point?

16 What would the three angles at this point add up to?

17 What is the size of each angle of a regular hexagon?

18 Can you explain why an equilateral triangle, square and regular hexagon will tessellate?

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Selected answers

1—Angles — basic properties

A **1** acute **2** obtuse **3** acute **4** right
5 reflex **6** straight **7** obtuse **8** acute

B **1** yes **2** yes **3** no **4** yes **5** 30°
6 35° **7** 15° **8** 58° **9** yes **10** yes
11 no **12** yes **13** 80° **14** 160°
15 85° **16** 62° **17** 65° **18** 38° **19** 45°
20 52° **21** 40° **22** 102° **23** 115°
24 50° **25** 40° **26** 100° **27** 155°
28 35° **29** 145° **30** 115° **31** 90° **32** 95°

C **1** 60° **2** 22.5° **3** 65° **4** 36° **5** 105°
6 41° **7** 24° **8** 33°

2—Angles and parallel lines

A **1b** The alternate angles are the same size
2b The corresponding angles are the same size
3b Co-interior angles add to 180°

B **1** 70° **2** 100° **3** 65° **4** 75°
5 105° **6** 116° **7** 78° **8** 85°
9 125° **10** 75° **11** 85° **12** 57°
13 69° **14** 100° **15** 48° **16** 82°
17 100° **18** 36° **19** 104° **20** 114°

C **1** parallel, a pair of alternate angles are equal
2 parallel, a pair of corresponding angles are equal
3 *not* parallel, the co-interior angles add to 190° (not 180°)
4 *not* parallel, the corresponding angles are not equal

3—Classifying triangles

A **1** acute-angled **2** right-angled **3** acute-angled
4 obtuse-angled **5** right-angled **6** acute-angled
7 obtuse-angled **8** obtuse-angled

B **1** isosceles **2** scalene **3** equilateral
4 isosceles **5** scalene **6** equilateral
7 isosceles **8** scalene **9** isosceles
10 equilateral **11** isosceles **12** scalene

C **1** acute-angled scalene **2** obtuse-angled scalene
3 right-angled scalene **4** right-angled isosceles
5 acute-angled isosceles **6** acute-angled equilateral
7 obtuse-angled scalene **8** right-angled isosceles
9 acute-angled isosceles **10** right-angled scalene
11 obtuse-angled isosceles **12** acute-angled equilateral
14 60° **15** 45° **16** No, the two obtuse angles would add to more than 180° (the sum of the 3 angles in a triangle)

4—Angle properties of triangles

A **1** right-angled **2** acute-angled **3** obtuse-angled
4 obtuse-angled **5** acute-angled **6** obtuse-angled
7 right-angled **8** acute-angled

B **1** 50° **2** 75° **3** 60° **4** 66° **5** 25°
6 40° **7** 22° **8** 42° **9** 120° **10** 125°
11 111° **12** 131° **13** 84° **14** 125° **15** 125°
16 109° **17** 142° **18** 63° **19** 33° **20** 140°
21 57° **22** 141° **23** 131° **24** 25°

C **1** $a = 115°$ **2** $x = 48°$ **3** $x = 70°$ **4** $x = 60°$, $y = 60°$
5 $x = 70°$, $y = 70°$ **6** $x = 75°$, $y = 30°$ **7** $a = 72°$
8 $y = 36°$ **9** $a = 63°$ **10** $x = 75°$, $y = 40°$
11 $a = 120°$, $b = 140°$, $c = 100°$ **12** $x = 75°$, $y = 45°$
13 $x = 65°$, $y = 50°$ **14** $x = 95°$, $y = 25°$
15 $a = 60°$, $b = 65°$ **16** $x = 125°$, $y = 55°$
17 $x = 30°$ **18** $x = 26°$ **19** $x = 20°$ **20** $x = 19°$

5—Quadrilaterals

A **1** square **2** parallelogram **3** rectangle
4 kite **5** trapezium **6** rhombus **7** trapezium
8 parallelogram **9** kite **10** rectangle
11 parallelogram **12** square

B **1** 2 **2** 4 **3** 0 **4** 1 **5** 2 **6** 0
7 2 **8** 4 **9** 2 **10** 0 **11** 2 **12** 0

C **1** 97° **2** 60° **3** 50° **4** 65° **5** 90°
6 85° **7** 42° **8** 49° **9** 125° **10** 45°
11 60° **12** 182°

6—Properties of a parallelogram

A **3** yes **5** yes **7** no **8** yes **9** yes
10 no **11** no **12** no **13** two, opposite, same, different, bisect

B **1** $x = 130°$, $y = 50°$ **2** $a = 116°$, $b = 64°$ **3** $x = 120°$, $y = 60°$ **4** $a = 46°$, $b = 134°$, $c = 46°$ **5** $x = 105°$, $y = 75°$ **6** $x = 40°$ **7** $a = 35°$ **8** $x = 42°$
9 $y = 51°$

C **1** $x = 60°$ **2** $x = 80°$ **3** $a = 58°$, $b = 52°$, $c = 70°$
4 $x = 59°$ **5** $x = 55°$ **6** $x = 23°$, $y = 42°$

7—Properties of a rhombus

A **4** yes **6** no **7** yes **8** yes **9** $\angle AEB = 90°$
10 yes **12** yes **13** $\angle KNL = 90°$ **14** yes
15 yes **16** all, parallel, same, lengths, right, diagonals, angles

B **1** $a = 105°$, $b = 75°$ **2** $x = 116°$ **3** $a = 115°$, $b = 65°$
4 $x = 55°$ **5** $x = 62°$ **6** $x = 90°$ **7** $x = 65°$, $y = 25°$
8 $a = 35°$, $b = 55°$ **9** $x = 63°$, $y = 54°$

C **1** $x = 45°$ **2** $a = 72°$, $b = 54°$, $c = 54°$
3 $x = 64°$, $y = 52°$ **4** $x = 95°$ **5** $x = 35°$
6 $a = 34°$, $b = 34°$, $c = 56°$

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8—Properties of a rectangle

A **2** yes **3** yes **5** yes **6** yes, $AE = EC$ and $BE = ED$
7 in half **8** no **9** opposite, parallel, right, length, diagonals

B **1** $x = 90°$ **2** $a = 62°$, $b = 28°$ **3** $x = 55°$, $y = 35°$
4 $a = 63°$, $b = 27°$, $c = 27°$ **5** $x = 40°$, $y = 50°$
6 $a = 42°$, $b = 84°$, $c = 96°$

9—Properties of a square

A **1** yes **3** yes **4** yes, $PT = TR$ and $ST = TQ$
5 in half **6** $\angle PTQ = 90°$ **7** right-angles
8 both 45°, yes **9** both 45°, yes **10** yes
11 same, parallel, right, length, bisect, diagonals, angles

10—Polygons

A **1** triangle **2** pentagon **3** quadrilateral
4 hexagon **5** octagon **6** quadrilateral
7 heptagon **8** pentagon

B **2**

Polygon	Number of sides	Number of triangles	Angle sum
Hexagon	6	4	$4 \times 180° = 720°$
Heptagon	7	5	$5 \times 180° = 900°$
Octagon	8	6	$6 \times 180° = 1080°$
Nonagon	9	7	$7 \times 180° = 1260°$
Decagon	10	8	$8 \times 180° = 1440°$

3 two less than the number of sides
4 $(n - 2) \times 180°$ **5** **a** 1260° **b** 2340° **c** 2520°
d 1620° **e** 3240° **f** 2160° **g** 2880° **h** 4140°
6 **a** 108° **b** 120° **c** 135° **d** 144° **e** 140° **f** 150°

C **1** 360° **2** **a** 108° **b** 72° **c** 360° **3** **a** 120°
b 60° **c** 360° **4** **a** 45° **b** 40° **c** 36° **d** 30°

11—Review — properties of plane shapes

A **1** isosceles **2** scalene **3** equilateral
4 scalene **5** square **6** parallelogram
7 kite **8** trapezium **9** pentagon
10 hexagon **11** quadrilateral **12** octagon

B **1** 55° **2** 55° **3** 45° **4** 77° **5** 75°
6 132° **7** 131° **8** 122° **9** 50° **10** 47°
11 115° **12** 119° **17** $x = 55°$, $y = 125°$
18 $a = 110°$, $b = 70°$ **19** $y = 73°$ **20** $x = 58°$
21 $x = 53°$, $y = 37°$ **22** $x = 65°$ $y = 25°$ **23** 720°
24 1440° **25** 1800° **26** 108° **27** 135°
28 144°

C **1** $a = 48°$ **2** $x = 62°$, $y = 56°$ **3** $x = 45°$, $y = 65°$
4 $x = 102°$ **5** $x = 63°$ **6** $x = 174°$
7 $x = 36°$ **8** $x = 95°$ **9** $x = 75°$

12—Translation

A **1** yes **2** no **3** no **4** yes **5** no **6** yes

C **2** yes **4** yes **5** yes **6** yes
7 distance, image, size, figure, congruent, vertices

13—Reflection

A **1** yes **2** yes **3** no **4** yes **5** no **6** no

C **1** yes **4** no **5** distance, same, angle, image, congruent, rotational

14—Rotation

A **1** yes **2** no **3** yes **4** yes **5** no **6** yes

C **1** image, length, same, area, figure, unchanged

15—Congruent figures

A **1** A and G, B and E, C and L, D and I, F and H, J and S, K and T, M and R, N and P, O and Q
2 A and H, B and D, C and F, E and G

B **1** **a** yes **b** yes **c** yes **2** **a** yes **b** yes **c** no
3 **a** yes, $AB = DF$, $AC = DE$ and $BC = EF$ **b** yes **c** yes

C **1** Yes, all sides should be 6 cm long, all angles should measure 90°, all the squares should fit exactly on top of each other (superimpose)
2 Yes, all the rectangles should superimpose
3 Yes, all the triangles should superimpose
4 No, the sizes of the angles were not specified

16—Congruent triangles

A A and E, B and F, C and J, D and H, G and I

B **1** SSS **2** SAS **3** SAS **4** AAS **5** RHS
6 SSS **7** AAS **8** SAS **9** SAS **10** **a** DE
b DF **c** EF **d** $\angle EDF$ **e** $\angle DEF$ **f** $\angle EFD$

C **1** yes, AAS **2** no **3** no **4** no **5** yes, AAS
6 yes, AAS

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17—Writing congruence proofs

A **1** SSS **2** SAS **3** RHS **4** AAS **5** AAS
6 a RP **b** PQ **c** QR **d** $\angle PRQ$ **e** $\angle PQR$ **f** $\angle RPQ$

B **1** $AB = FE$ (both 11cm)
$BC = ED$ (both 12 cm)
$AC = DF$ (both 8 cm)
$\therefore \Delta ABC \equiv \Delta DEF$ (SSS test)
3 $\angle RTS = \angle XYZ$ (both 90°)
$RS = XZ$ (both 18 cm)
$RT = YZ$ (both 11cm)
$\therefore \Delta RST \equiv \Delta XYZ$ (RHS test)
5 $\angle ACD = \angle ACB$ (both 90°)
$AD = AB$ (both 16 cm)
AC is common
$\therefore \Delta ADC \equiv \Delta ABC$ (RHS test)
6 $\angle ABC = \angle DEC$ (both 50°)
$\angle ACB = \angle DCE$ (vertically opposite angles)
$AC = DC$ (given)
$\therefore \Delta ABC \equiv \Delta CDE$ (AAS test)

18—Applying congruence

A **1** A and F (RHS test), B and H (AAS test), C and D (SSS test), E and G (SAS test)

B **1** $x = 9$ cm **2** $x = 9$ cm **3** $x = 7$ cm
4 $x = 10$ cm **5** $x = 37°$ **6** $x = 63°$ **7** $x = 16$ cm
8 $x = 17$ cm **9** $x = 18$ cm **10** $x = 60°$, $y = 40°$
11 $x = 60°$, $y = 25°$ **12** $x = 5$ cm **13** $x = 2$ cm
14 $x = 5$ cm **15** $y = 4$ cm

19—Geometric proofs

A **2** In the Δs PQR and PSR,
$\angle QPR = \angle SRP$ (equal alternate angles)
$\angle PRQ = \angle RPS$ (equal alternate angles)
PR is common
$\therefore \Delta PQR \equiv \Delta PSR$ (AAS test)
$\therefore PQ = RS$ and $PS = RQ$
(matching sides of the congruent triangles)
4 In the triangles AED and AEB,
$AD = AB$ (given)
$\angle DAE = \angle BAE$ (AC bisects the angle of the rhombus)
AE is common
$\therefore \Delta AED \equiv \Delta AEB$ (SAS test)
$\therefore \angle AED = \angle AEB$ (matching angles of the congruent triangles)
however $\angle AED + \angle AEB = 180°$ (BD is a straight line)
$\therefore \angle AEB = 90°$

B **3 c** The diagonals of a rhombus bisect the angles of the rhombus. **4 c** In an isosceles triangle, the angles opposite the equal sides are the same size.

20—Review — Congruent figures

A **1** A and F, B and D, C and H, E and G **2** A and H, B and D, C and F, E and G **3 a** DF **b** DE **c** EF
d $\angle DFE$ **e** $\angle FDE$ **f** $\angle DEF$

B **1** RHS **2** SSS **3** SAS **4** AAS **5** 10 cm
6 50° **7** 9 cm **8** 65°
9 In the triangles ABC and LMN
$\angle ACB = \angle LMN$ (both 55°)
$\angle BAC = \angle MLN$ (both 72°)
$BC = NM$ (both 12 cm)
$\therefore \Delta ABC \equiv \Delta LMN$ (AAS test)
10 In the triangles PQR and PSR,
$PQ = PS$ (both 18cm)
$\angle QPR = \angle SPR$ (given)
PR is common
$\therefore \Delta PQR \equiv \Delta PSR$ (SAS test)

21—Tessellations

C **1** yes **2** yes **3** no **4** yes **5** no **6** yes
7 yes **8** no **9** 60° **10** yes **11** 360°
12 square **13** yes **14** 360° **15** yes
16 360° **17** 120°
18 In order to tessellate, the angle size of the regular polygon must divide evenly into 360°.

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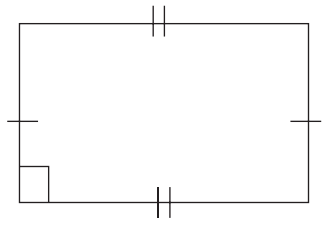
Properties of quadrilaterals – A summary

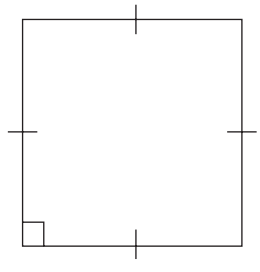
A *rectangle*

- has 2 pairs of parallel sides
- opposite sides are the same length
- all 4 angles are right-angles
- the diagonals are the same length
- the diagonals bisect each other
- has 2 axes of symmetry (through the midpoints of the sides)



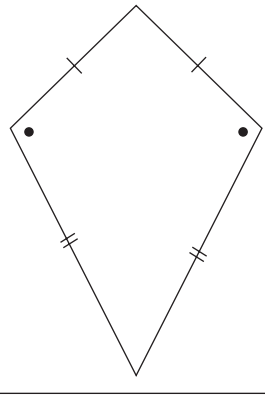
A *square*

- has 2 pairs of parallel sides
- all 4 sides are the same length
- all 4 angles are right-angles
- the diagonals are the same length
- the diagonals meet at right-angles
- the diagonals bisect each other
- the diagonals bisect the angles of the square
- has 4 axes of symmetry



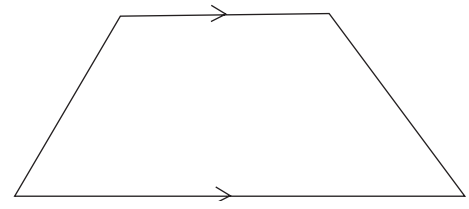
A *kite*

- has 2 pairs of equal sides
- one pair of opposite angles are the same size
- the diagonals meet at right-angles
- has only 1 axis of symmetry



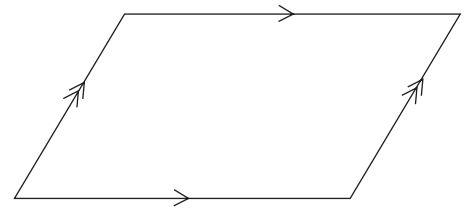
A *trapezium*

- has at least 1 pair of parallel sides



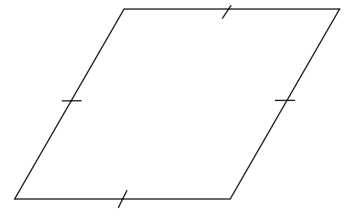
A *parallelogram*

- has 2 pairs of parallel sides
- opposite sides are the same length
- opposite angles are the same size
- the diagonals bisect each other
- has *no* axes of symmetry



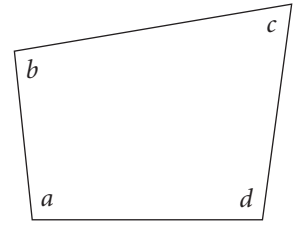
A *rhombus*

- has 2 pairs of parallel sides
- all 4 sides are the same length
- opposite angles are the same size
- the diagonals meet at right-angles
- the diagonals bisect each other
- the diagonals bisect the angles of the rhombus
- has 2 axes of symmetry (along the diagonals)



The four angles of *any* quadrilateral add to 360°.

$a + b + c + d = 360°$



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